Marginal Validity Is Not Enough: Reliability of Conformal Uncertainty Quantification for Mortgage Delinquency Screening

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Abstract

This thesis examines whether conformal prediction provides reliable uncertainty quantification for mortgage delinquency screening under severe class imbalance and temporal distribution shift. Using 2.322 billion loan-month observations from 46.65 million Freddie Mac loans from 1999 to 2023, a LightGBM model with isotonic probability calibration is evaluated chronologically across a calibration period and four subsequent economic regimes. Five conformal methods are compared on marginal, class-conditional, and temporal coverage, with subgroup and pocket-level audits added for the three static methods, alongside drift and operational screening measures. The results show that nominal marginal coverage can conceal severe conditional failure. At a 90% target, split conformal prediction achieved 90.30% marginal coverage on the calibration data but only 0.0039% coverage for delinquent observations. The failure is already present at the calibration anchor and later regime changes alter its severity. Adaptive prediction sets improved minority-class and subgroup reliability, whereas Mondrian calibration mainly reduced subgroup disparities and online-adaptive methods did not restore nominal coverage. Population stability monitoring did not reliably track these failures. At matched review budgets, split conformal prediction and adaptive prediction sets selected exactly the same loan-months as probability thresholding, while Mondrian routing captured fewer delinquencies. Overall, marginal conformal validity is insufficient for imbalanced, shifting screening systems. The findings rule out a ranking advantage for the monotone conformal rules studied here and instead point to a potential role for conformal prediction as a structured interface for abstention and conditional reliability auditing.

1 Introduction

1.1 Motivation

A real-estate credit-risk model's output is a probability that a loan will fall seriously delinquent within a fixed forward window. This is the input to a screening pipeline that must turn it into action. Decisions like these increasingly rest on machine-learning scorers, which are built for predictive accuracy rather than for the stable coefficient estimates through which a scorecard is read (Mullainathan & Spiess, 2017) and now match or outperform the logistic-regression scorecard (Lessmann et al., 2015), mortgage default included (Fitzpatrick & Mues, 2016). The decisions that matter are the rare ones with asymmetric costs: clearing a loan that then falls delinquent is a worse mistake than reviewing one that would have performed (Elkan, 2001). Acting on predictions therefore requires quantified confidence.

The tools credit-risk governance uses to assess this confidence were developed for related but different purposes. Discrimination metrics measure whether the model ranks delinquent loans above performing ones, calibration diagnostics ask whether its probabilities match observed frequencies and population-stability measures track score-distribution drift (Siddiqi, 2017; Yurdakul & Naranjo, 2020). However, none answers the question that is important for a screening decision: does the true outcome lie inside the model's stated confidence set, at the stated rate? A model can rank well, be calibrated on average and show no drift while never stating a set within which the truth is guaranteed to lie at a known rate.

Conformal prediction (CP) offers a formal marginal coverage guarantee for such sets. When wrapped around any base model, it returns not a point prediction but a label set for each loan, with a pre-specified marginal coverage rate - finite-sample, distribution-free, and model-agnostic (Angelopoulos & Bates, 2023; Vovk et al., 2005). The appeal for governance is immediate: the guarantee does not require the base model to be correctly specified, while attaching a formal coverage target to set-valued predictions. At first glance it is the reliability guarantee a screening pipeline lacks. The guarantee is not unconditional, however. It rests on exchangeability: the calibration data must be statistically interchangeable with the data encountered in deployment. A mortgage panel violates that structurally. The joint distribution of outcomes and borrower characteristics shifts across macroeconomic regimes (Campbell & Cocco, 2015), driven by aggregate conditions that loan-level features encode only partially (Deng et al., 2000). Over the 1999-2023 span of the Freddie Mac Single-Family Loan-Level Dataset (SFLLD), the delinquency rate varies across regimes by a factor of nearly two and a half. The same loan also appears across many serially dependent monthly observations.

Together, these break the interchangeability the guarantee assumes, most severely in the crisis conditions under which a screen matters most. Rarity works differently: at a screened-event rate of one to three percent of loan-months, class imbalance is not itself an exchangeability violation, but it lets the aggregate guarantee be carried by the majority class and say nothing about the minority the screen is meant to detect. These two failures have different origins, and that distinction organises the thesis.

How CP behaves once exchangeability fails has been studied theoretically (Foygel Barber et al., 2023; Gibbs & Candès, 2021). This thesis examines empirically how those failures present when the assumption is violated across multiple dimensions simultaneously. A further question is whether the aggregate guarantee is the appropriate objective, given that an average rate may hold steady while coverage collapses in the tail - specifically on the high-risk, minority-class loan-months the screening process exists to identify. This thesis therefore asks what happens to CP's outputs in a financially consequential, class-imbalanced, temporally non-stationary mortgage panel: through which mechanisms the guarantee becomes misleading, whether standard monitoring detects the failure, and whether the prediction sets change the decisions a screen would otherwise make.

1.2 Research Questions

That inquiry resolves into four research questions that move from the nature of the failure to its consequences for a deployed system. Each is answered in a dedicated section of Chapter 4.

The first concerns the failure itself: In a class-imbalanced loan-month panel with temporal regime shifts, through which mechanisms does the marginal coverage guarantee become misleading, and does the failure decompose into distinct kinds that a taxonomy can separate, rather than merely register?

The second concerns mitigation: Beyond the standard split baseline, the conformal literature offers Mondrian conditioning, adaptive prediction sets, and online-adaptive procedures, each targeting a different source or dimension of reliability failure. Across marginal, class-conditional, and temporal coverage, with subgroup and pocket-level audits for the static methods, how do these methods behave, and which sources of failure, if any, does each address?

The third concerns detection: A deployed system can monitor distributional change with the population stability index. How well does that standard diagnostic explain, or fail to explain, the reliability failures the first question identifies, and which coverage-specific diagnostics would a monitoring framework need in its place?

The fourth concerns the consequences: One test of a reliability guarantee is whether it actually alters the way the selection process operates. Does CP's statistical reliability translate into more efficient operational selection, for instance by identifying more delinquencies within a fixed review budget than would be the case with a simple ranking based on calibrated probabilities? Or is its value, where it has any, semantic and governance-related rather than selection-efficient?

1.3 Approach and Thesis Map

The empirical work uses the Freddie Mac SFLLD, a panel of 2.322 billion loan-months drawn from 46.65 million distinct loans between 1999 and 2023. A gradient-boosted model produces delinquency probabilities, recalibrated and passed to the conformal layer. The panel is split chronologically into a training window, a fixed calibration reference, and four sequential test windows. Each window is evaluated against a threshold frozen on an earlier period, the two online-adaptive procedures aside, so that coverage movement reflects distributional change. Five conformal methods are evaluated: the split baseline, Mondrian conditioning on a borrower-risk partition, adaptive prediction sets, and two online-adaptive procedures. All five are compared marginally, class-conditionally and temporally. The three static methods additionally undergo subgroup and pocket-level audits and operational translation. An isotonic-calibrated logistic regression is carried through the same pipeline as a robustness benchmark for the three static methods.

The thesis makes five contributions, summarised in the conclusion. The first gives it its title: marginal coverage can be preserved or even inflated while class-conditional, subgroup, pocket-level, and temporal coverage fail at once, a pattern the two-level taxonomy separates into an intrinsic, imbalance-driven failure and a distributional, regime-driven one. The second dissociates population stability from CP reliability: the standard monitor can rate the least reliable regime the most stable, masking exactly where coverage fails worst. The remaining three concern the equal-budget selection identity, the abstention-routing reframe - screening that operates through withheld decisions rather than positive flags - and a certification principle for deployment.

The thesis proceeds as follows. Chapter 2 develops the conceptual foundation; Chapter 3 specifies the data, methods, and evaluation framework; Chapter 4 reports the results, one section per research question; Chapter 5 draws out the implications for CP, credit-risk modelling, and governance; and Chapter 6 concludes.

Three disclaimers bound the claims. The first is that this is not an argument that CP is invalid: the guarantee holds wherever exchangeability does, and what I study is how its outputs behave where that condition fails. A question about a setting, not about the method's correctness. The second is that the thesis proposes no new conformal method, but evaluates existing ones and characterises their behaviour. The third is that it does not claim CP is valueless here, only that whatever value it has rests on grounds other than selecting loans more efficiently than the calibrated probability.

2 Background and Conceptual Foundation

This chapter introduces the concepts underlying the empirical analysis. §2.1 and §2.2 stay general: the conformal coverage guarantee, the exchangeability condition it rests on, and the impossibility of the stronger conditional guarantee a screening system would prefer. §2.3 to §2.6 turn to the setting, moving from the ways a mortgage panel violates exchangeability, through the two-level taxonomy organising the resulting failures, to the monitoring and decision layers above them. §2.2 and §2.3 together set up the second research question, §2.4 the first, §2.5 the third and §2.6 the fourth.

2.1 Conformal Prediction: Foundations and Marginal Coverage

CP turns the point output of any fitted model into a set of labels that contains the truth with a preset frequency. I use its split, or inductive, variant throughout, introduced by Papadopoulos et al. (2002) as a computationally efficient alternative to the original transductive construction, and now the standard form of the method (Angelopoulos & Bates, 2023). A base model is fitted on a training partition and thereafter held fixed. A disjoint calibration set of n labelled observations is passed through a nonconformity score $s(x, y)$ measuring how poorly the pair (x, y) conforms to the fitted model. The scores $s_1, \ldots, s_n$ are reduced to a single threshold $\hat{q}$, their $\lceil (n+1)(1-\alpha) \rceil / n$ empirical quantile, with α the user-chosen miscoverage level. For a test input x the prediction set collects every candidate label whose score does not exceed that threshold,

$$C(x) = \{y : s(x, y) \leq \hat{q}\}. \tag{2.1}$$

This threshold delivers a finite-sample coverage guarantee. For a test point drawn exchangeably with the calibration data,

$$1 - \alpha \leq \mathbb{P}\big(Y_{n+1} \in C(X_{n+1})\big) \leq 1 - \alpha + \frac{1}{n+1}, \tag{2.2}$$

where the upper bound holds when the score distribution is continuous, so coverage is not merely at least $1-\alpha$ but almost exactly $1-\alpha$ (Angelopoulos & Bates, 2023; Lei et al., 2018). Exchangeability is invariance of the joint distribution under any reordering of the observations, a condition weaker than independent and identical sampling. It makes the rank of the test score among the calibration scores uniform on $\{1, \ldots, n+1\}$, so that score falls at or below the empirical quantile with probability at least $(1 - \alpha)$. This is precisely the event that the true label is covered (Lei et al., 2018; Vovk et al., 2005). No parametric model and no assumption of correct specification enter: the guarantee is in this sense distribution-free and model-agnostic. Appendix A.1 states the formal results and the conditions each requires.

Two properties of this guarantee matter for what follows. Validity is a property of the calibration-to-test relationship, not of the model or the score: any score function yields the guarantee, and the score governs efficiency - the size of the resulting sets - and how coverage is distributed across classes and regions, never whether the guarantee holds (Angelopoulos & Bates, 2023). The guarantee is also marginal, the probability being averaged over the joint draw of calibration and test data (Lei et al., 2018), so it makes no claim about coverage at a fixed input, within a subpopulation, or within a deployment period. This is the silence this thesis documents.

2.2 Conditional Coverage and Its Impossibility

The natural repair is conditional coverage,

$$\mathbb{P}\big(Y_{n+1} \in C(X_{n+1}) \mid X_{n+1} = x\big) \geq 1 - \alpha, \tag{2.3}$$

which asks the guarantee to hold at each feature value rather than on average. Conditional coverage is strictly stronger, implying both marginal coverage and coverage within any sub-population. It cannot, however, be had. No distribution-free procedure achieves it in finite samples without making its sets useless: wherever the feature distribution places no atom at x, a conditionally valid set must be large enough to contain essentially every label (Foygel Barber et al., 2021; Lei & Wasserman, 2014; Vovk, 2012). Because a continuous feature assigns zero probability to each individual value, that condition holds almost everywhere, and the strengthening is out of reach for every conformal method.

The impossibility binds at individual feature values, and relaxing it to every subgroup above a probability floor gains nothing over a far stricter marginal level (Foygel Barber et al., 2021). What survives is a finite set of events fixed in advance; this thesis pursues two. The first is a region of feature space: Mondrian CP partitions the calibration data into pre-specified groups and computes a separate threshold within each. Coverage then holds within every group - local validity, sitting between marginal and conditional coverage (Lei & Wasserman, 2014; Vovk, 2012; Vovk et al., 2005) - at the cost of fragmenting the calibration sample across data-poor cells (Ding et al., 2023). The second is the label class itself. Class-conditional coverage,

$$\mathbb{P}\big(Y \in C(X) \mid Y = y\big) \geq 1 - \alpha \quad \text{for each class } y, \tag{2.4}$$

is therefore attainable distribution-free, and delivers the per-class control a marginal rate leaves unmonitored (Ding et al., 2023; Sadinle et al., 2019). Attaining it means a separate threshold within each class - the Mondrian construction with the label as its partition - which this thesis does not run (§3.5, §6.3). What it carries on this axis instead is adaptive prediction sets (APS), built for a different target - approximate coverage conditional on the features - and scoring a label through cumulative ranked probability mass rather than its own probability alone. Romano, Sesia, and Candès (2020) nonetheless report the score holding class-conditional coverage reasonably well without a per-class threshold.

Neither route recovers full conditional coverage: Mondrian secures validity only on the coarser partition it conditions on, APS carries no guarantee beyond the marginal rate, and each targets one failure axis. How far the two survive once the calibration-to-test relationship is itself non-exchangeable is part of research question 2, answered in §4.2; the third family it asks about is set out in §2.3.

2.3 Mortgage Panels as Multi-Layered Non-Exchangeability

A mortgage panel departs from the exchangeable setting of §2.1 through temporal drift and three structured dependence mechanisms, making it a demanding case for CP.

The first is temporal non-stationarity. The joint distribution of features and labels differs across the chronological windows into which the panel is divided: the positive rate alone varies across them by a factor of nearly two and a half (§3.1). More than the marginal rate moves. The conditional probability $P(Y \mid X)$ shifts too, because distress is driven by macroeconomic conditions - house-price-driven equity and the interest-rate environment - that loan-level features encode only partially; negative equity, the standard precondition for default, has its incidence set jointly by origination leverage and a subsequent aggregate price path that no borrower attribute determines (Campbell & Cocco, 2015; Deng et al., 2000; Foote et al., 2008). A threshold calibrated on one window's scores is therefore calibrated against a different joint distribution from the one it meets at test time: the simultaneous movement of $P(X)$ and $P(Y \mid X)$, rather than the more tractable case in which only the feature distribution changes (Sugiyama et al., 2007).

The remaining three are dependence structures rather than distributional movement. Repeated within-loan observations are the first: each loan contributes multiple rows over time, linked by scheduled amortisation and carried-forward delinquency history, so successive scores are serially dependent. Dependence alone need not imply non-exchangeability; here it compounds the calendar-time non-exchangeability already introduced above (Shumway, 2001). Overlapping labels are the second: because the label looks twelve months forward, a single distress event propagates the positive label to as many as twelve preceding loan-months of the same loan, so the apparent positive count overstates the number of independent credit events by roughly the window length. Cross-sectional common-factor dependence is the third: loans observed in the same calendar month share exposure to the same macroeconomic shock, correlating observations across loans within a period (Cameron & Miller, 2015).

Three responses in the conformal literature bear on this panel. Weighted CP restores the guarantee under covariate shift by reweighting calibration scores with the test-to-calibration likelihood ratio. But it holds $P(Y \mid X)$ fixed and lets only $P(X)$ move (Tibshirani et al., 2019), an assumption the regime shifts violate. The more general result bounds the coverage gap - the shortfall of realised coverage below nominal - by a weighted total-variation distance between the realised score sequence and its exchangeable (index-swapped) counterparts (Foygel Barber et al., 2023), so coverage degrades at most in proportion to the shift. The bound fixes neither the direction nor the magnitude of the loss for any particular shift and becomes vacuous once that distance is large. Adaptive conformal inference (ACI) and its extension relax exchangeability directly, adapting the set-forming level online from realised coverage (Gibbs & Candès, 2021, 2024); §3.5 specifies them. None restores exchangeability as each either bounds or adapts to its loss. Of the three, only the adaptive response is carried forward: ACI and DtACI make up the third of research question 2's method families, the other two being the conditional routes of §2.2. How each family behaves on its evaluated dimensions, and which source of failure it reaches, is answered in §4.2.

2.4 The Two-Level Failure Taxonomy

Two distinct things go wrong here, from unrelated causes. Class imbalance, which is not itself an exchangeability violation, can degrade conditional coverage on its own. The shift and dependence structures of §2.3 can perturb coverage separately. The intrinsic level is fixed at the calibration anchor and the distributional level is superimposed by calibration-to-test movement. Keeping them apart is what makes CP's behaviour here legible, since one shows as a coverage shortfall and the other can show as a surplus.

The intrinsic level (Level 1) arises from pooling different class-conditional score distributions under extreme imbalance. At a positive rate near one percent, the split-conformal threshold is the $(1-\alpha)$-quantile of a score distribution that is 99% negative-class, and so is, in effect, a quantile of the negative-class scores. The positive class occupies a different region of the score space, the more so because classifiers optimising aggregate performance under severe imbalance are dominated by the majority class (He & Garcia, 2009). Its scores therefore lie almost entirely above that threshold, and its true labels are rarely contained in the prediction set: split conformal prediction's known bias towards the majority class under imbalance (Löfström et al., 2015). The level chosen is incidental: the threshold stops being a quantile of the majority only once α falls to roughly the positive rate itself. How far above they lie is not settled by the ratio alone. The scale on which the scores are expressed fixes the distance the minority must cover, so a post-hoc calibration map compressing model outputs towards the base rate deepens the shortfall without changing the mechanism that produces it (§3.4; the map's incremental role is assessed in §4.1). That single threshold produces a second asymmetry across borrower segments, which differ in their positive rates: coverage is highest in the large, low-risk segments that supply most of the calibration mass, and lowest in the small, high-risk segments on which a screen concentrates. Class-conditional shortfall and subgroup heterogeneity are therefore two faces of one intrinsic mechanism. A third face follows: because coverage turns on where a score falls relative to one cut-point, failures concentrate in high-scoring regions that need not coincide with any partition fixed in advance. Failure at that finer, data-defined granularity is pocket-level coverage, measured in §3.6. The intrinsic level is, at bottom, the marginal guarantee's silence on class-conditional coverage and specialised to extreme imbalance. It would appear even under perfect exchangeability.

The distributional level (Level 2) is superimposed on this structure. The exchangeability violations explained in §2.3 do not create the vulnerability. They modulate its expression, moving the direction and size of the coverage deviation from one regime to the next. Four regime types organise that modulation, on two axes. Three follow the standard decomposition of dataset shift by where in the joint distribution the movement sits (Moreno-Torres et al., 2012), its loci borrowed without that scheme's causal-direction restriction and restated for the score distribution, from which the threshold is drawn rather than from the feature vector. A tail shift raises the positive rate sharply while the bulk of the score distribution barely moves - movement in the label margin $P(Y)$. A structural shift leaves the score distribution recognisable but distorts its mapping to the label, so that the observed positive class no longer stands for the same underlying event - movement in $P(Y \mid X)$. A composition shift changes the portfolio mix, compressing the score distribution without a proportional movement in the label - movement in $P(X)$. The fourth type separates on the second axis, abruptness: a stable drift is composition movement arriving gradually and without a break. Each type names the dominant movement, not the only one (§2.3), and is a claim about the character of the resulting deviation, but not yet its magnitude. §3.2 assigns calendar windows to the four types.

The two levels interact, and the pattern of this interaction is this thesis's object. Level 1 fixes the vulnerability - the minority class sits above a threshold set by the majority - and Level 2 governs whether a regime moves the deployment score distribution further from that frozen threshold or nearer. Neither movement can be read off the marginal rate, because of an elementary identity: writing π for the positive rate, marginal coverage is the class-conditional mixture

$$\mathbb{P}\big(Y \in C(X)\big) = (1-\pi)\,\mathbb{P}\big(Y \in C(X) \mid Y=0\big) + \pi\,\mathbb{P}\big(Y \in C(X) \mid Y=1\big), \tag{2.5}$$

each probability taken with respect to the realised calibration threshold, so at this panel's positive rates the minority term carries a weight of no more than three percentage points. The decomposition is the law of total probability. The objection to reading an average error rate as a per-category one is older (Vovk et al., 2005, §1.4). Over-coverage of the majority can therefore hold the marginal guarantee at or above nominal in every regime while coverage of the minority collapses. Marginal coverage preserved exactly where the minority-class failure is most severe, which is the precise sense in which it is not, on its own, a sufficient reliability criterion for screening. The two levels also organise the remedies already in hand: the class-conditional route of §2.2 is aimed at the class-conditional shortfall of Level 1, the Mondrian route at its subgroup heterogeneity, and the exchangeability-relaxing adaptive methods of §2.3 at Level 2. Nothing in the set is aimed at the pocket face, which is identified after the fact. Whether the two levels separate in practice, and whether their interaction takes the form set out here, is research question 1, answered in §4.1.

2.5 The Monitoring Gap: PSI and Conformal Reliability

A structured failure does not announce itself, so the question remains whether an existing monitoring apparatus would detect it. Model-risk-management guidance treats ongoing monitoring and outcomes analysis against realised results as complementary components of model validation (Board of Governors of the Federal Reserve System et al., 2026). One of the standard instruments is the population stability index (PSI), which compares the score distribution observed in deployment against the calibration-era reference and ordinarily needs no realised outcome to do it. Which score it is fed is a free choice and the score used here is evaluated at the realised label, information a live monitor would not have. Scores are binned, and writing p_b^{dep} and p_b^{cal} for the share of deployment and calibration observations falling in bin b, the index is

$$\text{PSI} = \sum_b \left(p_b^{\text{dep}} - p_b^{\text{cal}}\right) \ln \frac{p_b^{\text{dep}}}{p_b^{\text{cal}}}, \tag{2.6}$$

which is the standard credit-scoring convention (Siddiqi, 2017) and, for a fixed binning, the symmetrised Kullback-Leibler, or Jeffreys, divergence between the two distributions (Yurdakul & Naranjo, 2020). Industry practice reads a value below 0.10 as negligible, 0.10 to 0.25 as a moderate shift warranting investigation, and above 0.25 as a major shift requiring action (Siddiqi, 2017; Yurdakul & Naranjo, 2020). These are not calibrated cutoffs: at these sample sizes the asymptotic independent-multinomial reference of (Yurdakul & Naranjo, 2020) lies roughly four orders of magnitude below the 0.10 edge (Appendix D.3). Because the repeated loan-month panel does not satisfy that reference model, the index is reported here only as a descriptive magnitude.

The PSI and conformal coverage reliability, however, answer to different parts of the score distribution. Nothing forces them to agree. PSI's bins are the deciles of the calibration score (§3.6); the conformal threshold falls at the top edge of the ninth, so nine of its ten bins sit on the negative-class bulk below it. CP turns instead on the region above that threshold - where the minority class sits and where the subgroup heterogeneity is generated - which that binning resolves in a single bin.

Nothing in the construction of either fixes the sign of that relationship: a regime can register a large distributional shift while its coverage stays benign, or register almost none while its coverage fails. Whether they diverge on this panel, in which direction and by how much, is research question 3, answered in §4.3. Should they diverge, the statistic a governance function already trusts would misread the reliability state of a conformal screening system, and a coverage-aware monitor would have to track something other than movement in the score bulk. Neither a marginal coverage rate nor a bulk stability index would suffice on its own.

2.6 The Operational Question: Prediction Sets as Screening Decisions

For the operational analysis, the possible prediction sets are mapped here onto three actions: the negative singleton $\{0\}$ clears a loan automatically, the positive singleton $\{1\}$ flags it as high-risk, and the non-committal sets - the empty set and the two-label set $\{0, 1\}$ - route it to manual review. In this form a conformal rule is a reject-option classifier in the sense of Chow (1957, 1970). It commits where it is confident and abstains where it is not. Against a fixed cost of abstention Chow's rule reduces in the binary case to a pair of thresholds on the posterior probability (Herbei & Wegkamp, 2006), and the operating points it traces as the abstention rate varies form a risk-coverage curve (El-Yaniv & Wiener, 2010; Geifman & El-Yaniv, 2017). Here, coverage means the non-abstaining share, not the conformal sense used throughout.

Consider a conformal rule whose clearance region (the loans receiving the performing singleton) is a lower interval of a single quantity, here the model's estimated delinquency probability, cut at one threshold common to the whole portfolio and fixed over the period compared. At a matched review volume such a rule clears and reviews precisely the loans a threshold on the probability would. The equivalence is an identity argued here, and §4.4 confirms it numerically. What secures the equivalence is the threshold rather than the simplicity of the score, or the information the score reads: a rule setting a separate cut-point within each group of a partition clears a union of group-wise intervals, and a rule moving its cut-point between periods clears a different interval in each. Either departure reorders loans against the global probability and breaks the equivalence. Which methods satisfy both conditions, and which fail one, is settled in §3.5. Whether the conformal layer therefore adds operational value beyond ranking by that probability, and whether a method whose routing is not monotone in it behaves differently, are questions the reliability metrics of research questions 1-3 cannot reach. They are research question 4, in §4.4.

3 Data and Methodology

This chapter describes the empirical implementation and the criteria used to evaluate it. §3.1-§3.3 build the panel: the loan-month unit and label, the chronological split into four regimes, and the feature set. §3.4 fits and calibrates the base scorer and its logistic benchmark. §3.5 specifies the five conformal methods under comparison, and §3.6 the coverage, monitoring and operational measures applied to their output.

3.1 Dataset and Observation Unit

As stated, the empirical base is the Freddie Mac SFLLD (standard release), the standard public panel for mortgage credit research, covering fixed-rate single-family mortgages originated from 1 January 1999 onward (Freddie Mac, 2026b). Loans outside the Standard Dataset's eligible product scope lie beyond this panel. Appendix B.1 details the exclusions and treatment of super-conforming loans. I restrict the observation window to 1999 through 2023, so that the forward label window of every retained loan-month closes within the available performance history.

The unit of analysis is the loan-month: one row per active loan per calendar month, entering the base population only if the loan is current or 30-59 days delinquent and has reported loan age at least one. Loans already 60 days or more delinquent at t are excluded, since the target is the onset of distress from a performing or early-delinquency state, not the continuation of an already-distressed one. The design is a fixed-horizon discrete-time event-history panel (Singer & Willett, 1993), at the granularity at which a screening system acts.

The target is binary: the label of loan-month t is

$$y_t = \mathbf{1}\{\, \exists\, s \in (t,\, t+12] : D_s \geq 60 \ \text{ or } \ \mathrm{REO}_s = 1 \,\}\,, \tag{3.1}$$

where D_s is the days-past-due count and REO_s the real-estate-owned status at performance month s (Freddie Mac, 2026b). I call this event delinquency and reserve default for the economic concept: sixty days of arrears is an entry into distress, not a completed default, and no claim about ultimate loss follows from it. Forward-window information never enters the feature set (Kaufman et al., 2012), which establishes a row-level learn-predict separation. Two factors limit the accuracy of the classification. Code-01 payoff/maturity yields $y_t = 0$ unless a target event occurs first (Deng et al., 2000); a sensitivity analysis excluding affected rows is summarised in §4.2. A small share of loans default through credit-event disposition without ever recording an active 60-or-more-days-past-due state. These fast defaults escape the label and amount to 0.17% of the loans that carry a positive label (Appendix B.2), a bounded, one-sided undercount.

In the analysis panel the positive class is rare throughout, between 1.04% and 2.53% across windows (Table 3.1): a structural condition under which the intrinsic level of §2.4 operates. Four sampling schemes run through the analysis. Model fitting, conformal calibration and test-window coverage evaluation use full rows; the discrimination and score-level drift diagnostics use a two-percent subsample, the local reliability audit and operational routing analyses and the robustness benchmark of §3.4 five-percent ones, and the feature-level drift diagnostics of §3.6 a ten-percent one. All draws are deterministic on the loan identifier, so every monthly observation of a loan carries the same role, and the full allocation of rows is tabulated in Appendix B.3.

The panel is subject to the layered dependence of §2.3, and because the chronological design places successive windows of one long-lived loan into different splits, that within-loan correlation crosses split boundaries by construction.

3.2 Chronological Split and Regime Design

The panel is partitioned by observation month: a training window, a calibration window, then four sequential test windows, never shuffled across a boundary. Time-ordered evaluation is appropriate for out-of-period behaviour under distributional change (Roberts et al., 2017; Tashman, 2000). Here chronology refers to observation time, with the outcome-horizon qualification below.

The training window spans 1999-2004 with its final year reserved for early-stopping validation (§3.4); calibration spans 2005-2006 and serves as the conformal reference. Static thresholds are fixed here and applied unchanged later (§3.5), making it the calibration anchor for regime comparisons. The split is not purged by the twelve-month outcome horizon: 51.8% of calibration rows have label windows extending into 2007, so this is not a strict real-time deployment backtest. Their positive-rate difference is -0.02 percentage points against the clean-half counterfactual. Appendix B.2 does not identify the corresponding effect on the conformal threshold.

The four test windows are defined to span the four regime types of §2.4 (Table 3.1). The 2007-2012 crisis, with the delinquency aftermath that outlasts it (NBER recession December 2007 to June 2009; National Bureau of Economic Research, 2010), stands as a tail shift, the 2013-September 2019 recovery as a stable drift, the March 2020-September 2021 pandemic and forbearance period (NBER recession February to April 2020; Business Cycle Dating Committee, 2021) as a structural shift, and the 2022-2023 rate-hiking cycle, which over 2022 lifted the federal funds target range from 0-0.25% to 4.25-4.50% (Board of Governors of the Federal Reserve System, 2022; Kozlowski & Jordan-Wood, 2023), as a composition shift. I refer to the four throughout as the crisis, normal, COVID and rate-hike windows, the short names Table 3.1 carries. The mechanism labels are a priori design hypotheses, each naming the dominant observable shift rather than a proven cause. The evidence bearing on them - the stability index, the mean-score shift and the realised positive rate - is set out in §4.3 and per window in Appendices F.1 and F.2.

Table 3.1: Chronological split and regime design

Window	Period	Role	Loan-months (M)	Loans (M)	Pos. rate (%)
Training	1999–2004	Fit + early-stop	248.1	12.7	1.16
Calibration	2005–2006	Conformal reference	179.3	9.7	1.11
Crisis	2007–2012	Test	633.2	17.7	2.53
Normal	2013–Sep 2019	Test	753.7	18.6	1.52
COVID	Mar 2020–Sep 2021	Test	201.4	16.6	1.68
Rate hike	2022–2023	Test	306.3	15.0	1.04

Note. Distinct-loan counts are per window and do not sum to 46.65 million distinct loans, because a long-lived loan contributes observation months to several windows. Positive rates are the full-panel window composition.

The COVID window shows the sharpest version of this caveat. CARES Act forbearance let borrowers with federally backed mortgages suspend payments on attestation of hardship (Conference of State Bank Supervisors, 2020; Federal Housing Finance Agency Office of Inspector General, 2020), and the 2020 payment-deferral option moved missed payments into a non-interest-bearing deferred balance (Freddie Mac, 2026a, 2026b), so pandemic-era policy could alter how borrower hardship translated into the observed 60+ DPD/REO outcome. Buffer months are excluded at the normal/COVID and COVID/rate-hike transitions, but the overlap cannot be removed entirely: roughly 9% of the normal window retains COVID-tinged forward labels, carrying a positive rate of 3.2% against 1.3% in the remainder. The windows are aligned to regimes but not internally homogeneous, the crisis alone moving from a positive rate of 1.3% in January 2007 to 3.6% by December 2008 (§4.3). The rate-hike window carries a different qualifier: on a fixed-rate book a rate cycle moves who originates before it moves who defaults, and the twelve-month label closes by the end of 2024, so its 1.04% rate registers the composition change ahead of any slower credit response.

3.3 Features and Panel Construction

Each loan-month is described by 29 features in four families, documented in full in Appendix B.4: static origination characteristics, among them credit score, loan-to-value (LTV), debt-to-income (DTI), note rate and term, origination vintage entered categorically by year and quarter, and geography by census division (U.S. Census Bureau, 2010); dynamic current-state features measured at t, among them unpaid balance, the three-month relative balance change, the current note rate and a 30-day-delinquency indicator; a twelve-month delinquency history, of which recent arrears is the economically central predictor (Fitzpatrick & Mues, 2016); and binary missingness flags on the disclosure-subject origination fields.

Every feature is observable at or before t. Eight raw fields are hard-excluded and kept only as diagnostics, on two grounds. The modification, deferral, deferred-balance and workout fields record servicing intervention rather than borrower state, and only the first varies in the training window. Loan age, remaining maturity, the amortisation ratio and the origination channel are temporal proxies: the first three are strongly tied to seasoning and maturity, while granular broker and correspondent reporting begins only in 2008, so channel partly encodes disclosure era. A training-window correlation screen, reported in Appendix B.4, flags only the current and lagged 30-day-delinquency indicators, both legitimately observable at t.

Missing values are preserved rather than imputed: non-disclosure of credit score, LTV and DTI is cleaned from raw sentinels to explicit nulls paired with missingness flags, and the gradient-boosted learner routes nulls natively (§3.4), so disclosure status enters as a potential signal. Imputing the values instead would require an ignorability assumption (Rubin, 1976).

Credit score (= FICO in the thesis) and LTV also define the borrower-risk partition used throughout the reliability analysis. Tiers of the disclosed origination Credit Score (Prime ≥ 720, Near-Prime 640-719, Subprime < 640), labelled FICO tiers in the implementation, crossed with LTV buckets (Low $\leq 80\%$, Moderate 81-95%, High $> 95\%$), give nine cells plus a tenth sentinel cell for undisclosed credit score or LTV. Credit score is the disclosure's underwriting measure of borrower quality, and leverage the standard axis of mortgage-default risk (Campbell & Cocco, 2015; Foote et al., 2008). On the calibration window the cells span a wide risk range, monotone in each margin, from 0.25% in the Prime $\times$ Low cell to 9.29% in the Subprime $\times$ High cell, and they are severely unequal in size, the Prime $\times$ Low cell alone holding over half the calibration rows and the Subprime $\times$ High cell under a tenth of one percent of them (Appendix B.5). That range makes the partition a meaningful test of portfolio-wide reliability, and supplies the structure the Mondrian calibration of §3.5 and the subgroup metrics of §3.6 act on.

3.4 Base Model, Probability Calibration, and Robustness Benchmark

The base scorer is a LightGBM gradient-boosted decision-tree model (Friedman, 2001; Ke et al., 2017), chosen for its performance on credit data (§1.1) and for the sparsity-aware missing-value handling it shares with XGBoost (Chen & Guestrin, 2016) - a default direction learnt at each node - which consumes the non-disclosure encoding of §3.3 without imputation. It is fitted on the 1999-2003 training rows, with 2004 reserved for early-stopping validation (§3.2). Class imbalance is handled through `scale_pos_weight`, set to the negative-to-positive ratio on the fitting rows alone, roughly 80.6. That differs from the ratio implied by Table 3.1, which includes the 2004 early-stopping year and its lower positive rate. Hyperparameters are tuned by tree-structured Parzen search and the model refit at full scale to its early-stopping optimum of 646 trees, with the full configuration and a feature-attribution panel in Appendix C.1.

The fitted model discriminates well throughout the deployment period. The area under the receiver-operating-characteristic curve (AUROC) falls from 0.914 on the calibration window to between 0.826 and 0.883 across the test regimes, lowest in the COVID window (Appendix C.3). Average precision is below the calibration value in three of the four and the crisis-window exception reflects that window's doubled positive rate rather than better ranking, since precision moves with class skew where AUROC does not (Davis & Goadrich, 2006; Fawcett, 2006) and the crisis AUROC is the lower of the two. The reliability failures of §4 therefore arise on top of a competent scorer. One caveat attaches to the frozen specification: the share of observations from vintages absent from the 1999-2004 training split rises from 16% on the calibration window to 99% in the rate-hike window, so the model operates outside its training vintage support, and that binds the robustness benchmark below more tightly still.

The raw outputs are not calibrated probabilities: the re-weighted scores systematically overpredict, with a mean raw score of 0.158 on held-out calibration data against an observed positive rate of 0.011 and an expected calibration error over ten equal-width bins of 0.147 (Guo et al., 2017).

They are therefore recalibrated with isotonic regression via the pool-adjacent-violators algorithm (Ayer et al., 1955; Zadrozny & Elkan, 2002), standard post-hoc practice for tree ensembles (Niculescu-Mizil & Caruana, 2005). The calibrator is fitted once on a loan-level hash holdout of the calibration window, roughly a tenth of calibration loans, and applied unchanged to every test regime. The complementary nine residue classes form the conformal calibration partition of §3.5, so calibrator-fitting rows never influence conformal calibration. Recalibration restores probabilistic fidelity on held-out data - the out-of-sample Brier score (Brier, 1950) falls from 0.064 to 0.009 - while leaving the ranking essentially unchanged, AUROC moving less than 0.001 (Appendix C.2).

Conformal validity rests on the exchangeability of calibration and test scores, not on their correctness as probabilities (§2.1), so recalibration is not required for the coverage guarantee. It is adopted because the adaptive prediction-set score is built from the calibrated probability vector (§3.5), because the drift and operational analyses of §4.3-§4.4 are interpretable only on a probability scale, and because cross-model comparison against the logistic benchmark requires a common calibrated footing. This is not a neutral choice, as the isotonic map compresses the upper score range sharply towards the roughly one-percent base rate - a raw 0.90 maps to 0.14, a raw 0.50 to 0.018. How far the compression deepens the class-conditional coverage failure beyond the failure already present before recalibration is assessed in §4.1 on the benchmark pipeline, where the same construction can be run with and without the map. The contrast identifies the map's incremental role, but is not supposed to isolate the full mechanism experimentally.

A second scorer is carried through the pipeline end to end as a robustness benchmark, so that the reliability failures documented in §4 cannot be attributed to the gradient-boosted learner. It is an L_2-penalised logistic regression with balanced class weights - the scorecard form machine-learning scorers have come to match or outperform (§1.1) - fitted on a five-percent loan-level sample of the same 1999-2003 window and using the same 29 panel features under the benchmark-specific preprocessing of Appendix C.3. It has its own isotonic calibrator and conformal calibration partition, so no fitted component is shared, while the split boundaries, underlying feature definitions, α and metric definitions are held fixed. Its discrimination is marginally below the gradient-boosted model's on the calibration window, 0.909 against 0.914, with no test window separating the two by more than 0.009 AUROC in either direction. Two asymmetries qualify the comparison. It enters the origination vintage as a standardised numeric, so the unseen vintages noted above shift its log-odds by linear extrapolation - a plausible contributor to its uniformly higher stability index (§4.3; Appendix F.4). And only the three static methods of §3.5 are re-evaluated under it. The two online-adaptive procedures are run on the primary pipeline alone.

3.5 Conformal Prediction Methods

All methods operate within the split-conformal framework of §2.1. The target miscoverage level is $\alpha = 0.10$; at the roughly 1.6×10^8 calibration observations (the nine hash residue classes of §3.4, nine-tenths of the calibration window) the finite-sample correction to the quantile of §2.1 is negligible, of order 10^{-9}. Score granularity is not: the upper bound in (2.2) assumes a continuous score, whereas the isotonic map of §3.4 is constant on each pooled block, so the scores are heavily tied and the empirical quantile falls on an atom the inclusion rule admits whole. Coverage therefore sits a little above $1 - \alpha$ even on the calibration rows from which the threshold is drawn. It is a standing offset fixed by the score's granularity rather than by any distributional movement, and one not visible for the randomised APS score. As stated earlier, I evaluate five methods: a split-conformal baseline, two static responses aimed at the intrinsic level of §2.4, and two online-adaptive procedures aimed at the distributional level.

The selection follows a one-mechanism-per-method rule, making the comparison diagnostic, not a contest. Hybrids that would address two sources at once are out of scope on that principle. The label-conditional calibration of §2.2 (Vovk, 2012) is not excluded by it, being one mechanism, but is simply left unrun, and §6.3 takes both up; weighted CP is excluded on substance, for the reason set out in §2.3 (Tibshirani et al., 2019), and the beyond-exchangeability bounds of Foygel Barber et al. (2023) enter only as interpretation.

The baseline is split conformal prediction (SCP). Let $\hat{p}(x)$ denote the isotonically calibrated estimate of $\mathbb{P}(Y = 1 \mid X = x)$ from §3.4 and $\hat{p}_y(x)$ the calibrated probability it assigns to candidate label y. The SCP nonconformity score is one minus the calibrated probability of the candidate label,

$$s_{\text{SCP}}(x, y) = 1 - \hat{p}_y(x), \tag{3.2}$$

and the prediction set retains every label whose score is at or below the frozen global threshold $\hat{q}$ of §2.1 (Sadinle et al., 2019). The two labels therefore enter under mirrored conditions: the performing label where $\hat{p}(x) \leq \hat{q}$, the delinquent label where $\hat{p}(x) \geq 1 - \hat{q}$. Because $\hat{q}$ sits well below one half here, these cannot hold together, so SCP never returns the two-label set and every loan whose calibrated probability falls between them returns the empty set. And because recalibration compresses the upper score range (§3.4), the delinquent condition is almost never met, so SCP's response to a high-risk loan is abstention rather than a positive flag (§4.1). Its clearance region is accordingly the lower interval $\{x : \hat{p}(x) \leq \hat{q}\}$, on one threshold common to the portfolio and frozen throughout: SCP satisfies both conditions of §2.6 and orders loans as the calibrated probability does.

Mondrian conformal prediction, the first of the two routes of §2.2, fixes a separate threshold $\hat{q}_g$ within each group g of a pre-specified partition (Vovk, 2012; Vovk et al., 2005). I apply it primarily on the borrower-risk partition of §3.3, and additionally on FICO tier, LTV bucket and census division alone. Thresholds for all four partitions are in Appendix D.1. Its clearance region $\{x : \hat{p}(x) \leq \hat{q}_{g(x)}\}$ is therefore a union of cell-wise lower intervals and not one lower interval of the global probability: Mondrian fails the first condition of §2.6, and a loan cleared in one cell can carry a higher probability than one reviewed in another. What that reordering costs is measured in §4.4.

The APS keeps a single global threshold but replaces the score. For a candidate label y, let $\rho_y(x) = \sum_{y' : \hat{p}_{y'}(x) > \hat{p}_y(x)} \hat{p}_{y'}(x)$ be the calibrated probability mass on labels strictly more probable than y. The randomised APS score accumulates that mass and adds a uniformly weighted fraction of the candidate label's own probability,

$$s_{\text{APS}}(x, y, U) = \rho_y(x) + (1 - U)\, \hat{p}_y(x), \qquad U \sim \text{Uniform}(0, 1), \tag{3.3}$$

the generalised inverse-quantile construction, whose randomisation allows fractional inclusion of the boundary label (Romano, Sesia, & Candès, 2020). It is the second route of §2.2, redistributing coverage across the class-conditional margins through cumulative ranked probability mass. The randomisation is retained for the coverage analysis; the operational routing of §4.4 needs a reproducible disposition and uses the non-randomised reduction $U = 1$, which scores each label by the preceding mass $\rho_y(x)$ alone against $\hat{q}_{\text{APS}}$, the threshold calibrated from the randomised scores. No mass precedes the most probable label, so that label is never dropped and the reduction returns no empty set: the APS review queue is made of two-label sets where SCP's is made of empty ones. The other label joins it exactly when the more probable one's calibrated probability does not exceed $\hat{q}_{\text{APS}}$. Because that threshold, unlike SCP's, sits well above one half, clearance reduces to the lower interval $\{x : \hat{p}(x) < 1 - \hat{q}_{\text{APS}}\}$, review to the band up to $\hat{q}_{\text{APS}}$, and escalation to everything above. Despite that different score construction, APS satisfies both conditions of §2.6 as SCP does.

The remaining methods address temporal drift by relaxing exchangeability itself. ACI updates the working miscoverage level α_t online in response to realised coverage, so that $1 - \alpha_t$ is the effective target at which its sets are formed. Writing err_t for realised miscoverage at update step t,

$$\alpha_{t+1} = \alpha_t + \gamma\,(\alpha - \text{err}_t), \tag{3.4}$$

so the sets widen after undercoverage and narrow after overcoverage, with the step size γ governing each adjustment (Gibbs & Candès, 2021). Because γ trades adaptation speed against stability, I evaluate the grid $\gamma \in \{0.005, 0.01, 0.02, 0.05\}$. Dynamically-tuned ACI (DtACI) removes that choice by running $K = 500$ candidate step sizes as experts, each maintaining its own level α_t^i, and outputting their weighted combination

$$\alpha_t = \sum_{i=1}^{K} p_t^i\,\alpha_t^i, \tag{3.5}$$

where the weights p_t^i follow an exponential expert-aggregation rule configured in Appendix D.2 (Gibbs & Candès, 2024; Gradu et al., 2023). Both methods produce one global threshold per calendar month. For ACI, err_t in (3.4) is that month's realised miscoverage rate; for DtACI, each expert uses the corresponding monthly rate at its own threshold, and the updated expert states and weights determine the next weighted level in (3.5). All states start from $\alpha_0 = 0.10$ and evolve across the 196 observed months of the 2007-2023 test period. The buffer months excluded at the regime boundaries are absent from that sequence, so the working level carries across each break unchanged. A threshold that moves between periods fails the second condition of §2.6, which is why §4.4 holds both methods out of the operational comparison. One idealisation is carried with them: the update reads err_t from the realised labels of month t, which the twelve-month forward window of §3.1 does not close until $t + 12$, so a live system could feed these procedures only a year-old error signal. The evaluation instead uses contemporaneous labels and is therefore not a literal real-time backtest.

3.6 Evaluation Framework

Because a marginal rate can hold while conditional reliability fails along the axes of §2.4, I assess coverage at a hierarchy of granularities, then read the prediction sets as screening decisions. For any evaluation set G of loan-months, empirical coverage is the fraction whose prediction set contains the realised label,

$$\widehat{\mathrm{Cov}}(G) = \frac{1}{|G|} \sum_{i \in G} \mathbf{1}\{y_i \in C(x_i)\}, \tag{3.6}$$

with marginal coverage taking G as the full test window.

The finer measures apply (3.6) on narrower strata. Class-conditional coverage restricts G to each label in turn, separating reliability on the common performing class from the rare delinquent one. For the static methods, subgroup coverage restricts G to each FICO $\times$ LTV cell, summarised by the cell-level coverage gap (CovGap), here the largest absolute deviation from nominal across cells rather than the mean deviation across classes of Ding et al. (2023), and by worst-group coverage (WGC), the least-covered cell (Romano, Foygel Barber, et al., 2020). Temporal coverage evaluates (3.6) on a twelve-month rolling window, resolving within-regime dynamics that a per-regime average smooths away. The online-adaptive methods are additionally read at their long-run rate, pooled over all 196 months. Coverage is also profiled on deciles of the calibrated score, the common risk axis on which §2.4 expects failure to order. This is a diagnostic cut rather than a sixth reliability dimension.

Marginal, class-conditional and subgroup coverage are also computed on the calibration window itself, as the calibration anchor of §3.2. Taken on the rows from which the threshold is drawn, the marginal figure carries no information, sitting at nominal by construction up to the granularity offset of §3.5. The conditional ones do, since a failure already present there does not require calibration-to-test movement to explain it.

For the three primary static methods, the finest granularity is pocket-level coverage: whether miscoverage concentrates in identifiable regions. A gradient-boosted classifier predicting the miscoverage indicator $M_i = \mathbf{1}\{y_i \notin C(x_i)\}$ from covariates measures how predictably failure is associated with observable position (Braun et al., 2025; Zhou et al., 2026), fitted on the score alone and on the full feature set to isolate the calibrated score's role (Appendix D.3). One caveat attaches: under a randomised score the set is itself random, so miscoverage carries a component no covariate predicts, and the attainable discrimination is bounded below its deterministic-score counterpart. A low value there does not by itself certify uniform coverage. A concentration curve summarises how tightly that model localises failure: ranking observations by predicted miscoverage risk in decreasing order σ, it plots the share of realised miscoverage in the highest-risk fraction f,

$$\mathrm{Conc}(f) = \frac{\sum_{k=1}^{\lfloor fn \rfloor} M_{\sigma(k)}}{\sum_{i=1}^{n} M_i}, \tag{3.7}$$

with the concentration area $\int_0^1 \mathrm{Conc}(f)\, df$ running from $\frac{1}{2}$ under diffuse failure towards its ceiling of $1 - \frac{1}{2}\widehat{\mathbb{P}}(M = 1)$ under sharp concentration. Depth-three decision trees (Breiman et al., 1984) then surface the worst-covered pockets as interpretable regions (§4.1, §4.2).

Against these coverage measures I set the drift statistic a deployed system would already run. The population stability index (2.6) is computed over ten equal-frequency bins of the split-conformal nonconformity score, taking the calibration window as reference and each test window's two-percent subsample as comparison, so that it summarises movement in the same quantity from which the conformal threshold is drawn. That quantity is evaluated at the realised label, as the threshold itself is, so the index sees what a deployed monitor could not: the movement of the positive class's own scores. The choice runs in the incumbent statistic's favour, as a label-free index has strictly less of the information that would let it track coverage. The feature-level variant, computed against the same calibration reference on the origination characteristics over a ten-percent subsample and needing no outcome at all, supplies the label-free check and is reported alongside it in §4.3. Score-PSI bin construction is in Appendix D.3 and feature-level results are in Appendix F.3.

The prediction sets are then read as screening actions under the three-way disposition mapping of §2.6, and four quantities describe the resulting queues. The false auto-clear rate is the share of delinquent loan-months cleared automatically, $\widehat{\mathbb{P}}(\text{auto-clear} \mid Y = 1)$, and capture is its complement. Auto-clear contamination, $\widehat{\mathbb{P}}(Y = 1 \mid \text{auto-clear})$, conditions on the disposition, so a low contamination rate and a high false auto-clear rate coexist when positives are rare. Review precision, $\widehat{\mathbb{P}}(Y = 1 \mid \text{review})$, and review lift, its ratio to the portfolio positive rate, describe the review queue. Capture and class-conditional coverage can diverge sharply: capture requires only that a delinquent loan-month's set was not the performing singleton, coverage that it contained the delinquent label, so the empty set satisfies the first and fails the second.

Finally, the equal-review-budget capture gap isolates whether conformal routing captures more delinquencies than a probability cut-off at matched review cost, the budget being the share of loan-months a rule does not clear automatically (§2.6). Matching the budget makes this a bounded-abstention comparison at a fixed accepted fraction (Franc et al., 2023), holding the operating point constant so that any difference is attributable to the ordering of loans. At a review budget b,

$$\Delta_{\text{cap}}(b) = \text{Capture}_{\text{CP}}(b) - \text{Capture}_{\text{prob}}(b). \tag{3.8}$$

Every coverage deviation is judged against a practical-significance band of ± 2 percentage points. At sample sizes of order 10^8, with each loan contributing many serially correlated monthly observations under overlapping labels, a binomial interval treats them as independent and understates uncertainty, so essentially every deviation is nominally significant (Cameron & Miller, 2015). Clopper-Pearson intervals (Clopper & Pearson, 1934) are reported in Appendix D.3 for scale, but it is the band, not interval exclusion, that decides materiality.

4 Empirical Results

This chapter reports the empirical findings, taking the four research questions in turn. §4.1 establishes the joint coverage failure and reads it through the two-level taxonomy; §4.2 compares what the five conformal variants repair; §4.3 asks what a deployed drift monitor detects; and §4.4 reads the prediction sets as screening decisions.

4.1 Marginal Validity and the Failure Taxonomy (RQ1)

Marginal coverage holds throughout and describes almost nothing a screening system needs to know. On the conformal calibration partition, the rows from which the threshold is drawn, SCP attains 90.30% marginal coverage against the nominal $1 - \alpha = 0.90$. It covers the performing class at 91.31% and the delinquent class at 0.0039% (Table 4.1). At a positive rate of $\pi = 1.11\%$ the mixture identity (2.5) reconciles the three exactly, $0.9889 \times 91.31\% + 0.0111 \times 0.0039\% = 90.30\%$, the aggregate carried entirely by an over-covered majority. This is not a weak scorer: the base model separates the classes at an AUROC of 0.914 here (§3.4). Nor is it distributional movement: the 0.30 points above nominal are the score-granularity offset of §3.5, and the collapse would arise under perfect exchangeability, as the total-coverage oracle already exhibits it (Sadinle et al., 2019, §2.1). It is the marginal guarantee's indifference to class-conditional coverage (Ding et al., 2023; Löfström et al., 2015) at a positive rate near one percent, and the intrinsic level of §2.4.

The geometry of the nonconformity scores shows why (Figure 4.1). With the score distribution almost entirely performing-class mass, the threshold $\hat{q} = 0.0162$ is in effect a quantile of that class alone. One cut-point is applied to easy and hard loans (Romano, Sesia, & Candès, 2020): the mean nonconformity score is 0.009 for performing loan-months against 0.780 for delinquent ones, and all but 0.0039% of the latter score above it. The threshold sits so deep in the performing region that the delinquent label almost never enters a set: SCP emits essentially only the performing singleton or the empty set in every window, and 9.41 of its 9.70 percentage points of calibration miscoverage arrive as that empty set rather than a confident wrong answer. A device that clears or abstains rather than one that flags (§4.4). The blindness is constructional, not a property of the data. On the same calibration set the generalised inverse-quantile score of §3.5 covers the delinquent class at 37.60%.

The class ratio's contribution and the calibration map's can be told apart only on the benchmark pipeline, where the same construction runs with and without the isotonic map of §3.4. On uncalibrated logistic scores the threshold falls at 0.435 and split conformal covers the delinquent class at between 42.6% and 58.6% across the four regimes. The map moves the threshold to 0.018 and leaves that coverage between 0.14% and 1.43% (Appendix E.2). The class ratio makes the threshold a ma-

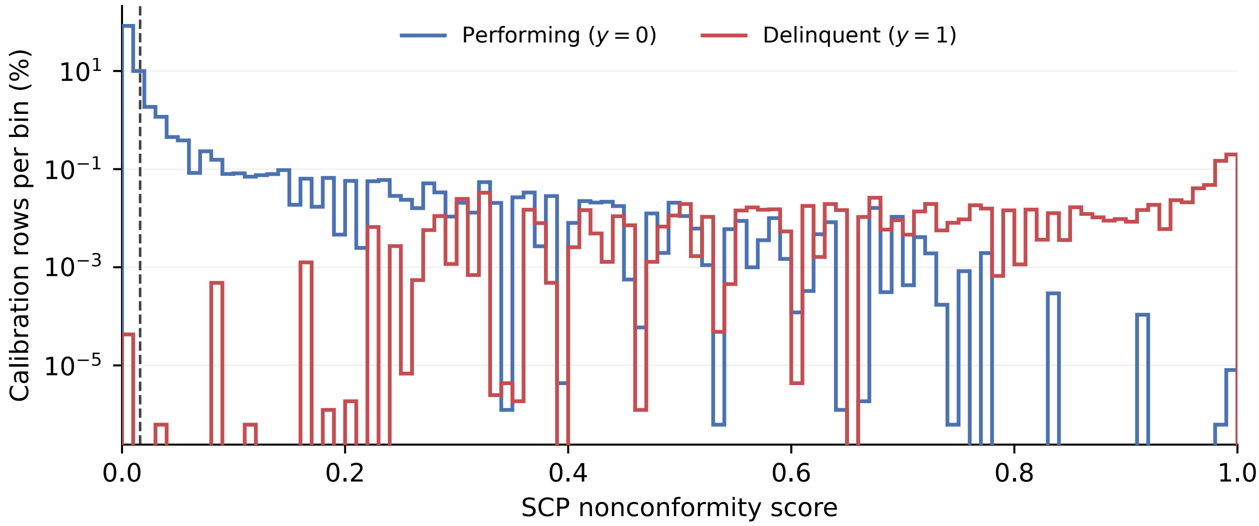

Figure 4.1: SCP nonconformity-score mass by true class on the conformal calibration partition; dashed line: $\hat{q}$.

jority quantile. The calibration map sets how far above it the minority sits, which is a mechanism identified on the benchmark scorer, not a magnitude measured on the primary one.

Across the test regimes the collapse deepens while the headline moves opposite to the reliability. Marginal coverage rises monotonically from 90.51% to 94.55%, 94.83% and 96.41% with delinquent-class coverage near zero throughout. What rises is performing-class coverage, from 91.31% at calibration to 97.42%, as more performing loan-months fall below the fixed threshold (Table 4.1). The 2007-2012 crisis is the instructive exception: its mean nonconformity score is the highest of the four, 65.8% above calibration, yet its marginal coverage is the lowest at 90.51% - that mean reflecting the window's doubled positive rate and a thin, score-heavy delinquent tail rather than any expansion of the performing bulk (§4.3). The crisis is accordingly the only test window inside the ± 2 percentage-point band of §3.6.

Table 4.1: Split conformal prediction reliability by regime

Window (mechanism)	MC (%)	Dev (pp)	$\widehat{\mathrm{Cov}}(y{=}0)$ (%)	$\widehat{\mathrm{Cov}}(y{=}1)$ (%)	Empty (%)	CovGap (pp)	WGC (%)	PSI
Calibration 2005–06 (anchor)	90.30	+0.30	91.31	0.0039	9.41	82.58	7.42	0.0000
Crisis 2007–12 (tail)	90.51	+0.51	92.86	0.0007	8.52	79.69	10.31	0.0062
Normal 2013–19 (stable)	94.55	+4.55	96.00	0.0002	4.82	47.07	42.93	0.0891
COVID 2020–21 (structural)	94.83	+4.83	96.45	0.0000	4.29	31.97	58.03	0.1240
Rate hike 2022–23 (composition)	96.41	+6.41	97.42	0.0001	3.08	24.94	65.06	0.1988

Note. Coverage columns use the full window rows; the calibration row uses the nine-tenths conformal partition from which the threshold is drawn (§3.4), and PSI the two-percent subsample (§3.1). PSI is zero at calibration, which is its own reference. CovGap and WGC span the ten FICO $\times$ LTV cells in Appendix E.1. MC = marginal coverage; Dev = deviation from nominal; Empty = empty-set rate.

The aggregate conceals more than the class split. Group-conditional coverage is the standard a marginal rate leaves unmonitored (Romano, Foygel Barber, et al., 2020), and on it miscoverage spans almost two orders of magnitude. Over the nine crossed FICO $\times$ LTV cells, calibration-window coverage runs from 99.03% in the largest, lowest-risk cell to 7.42% in the smallest, highest-risk one, the worst cell falling 82.58 points below nominal (Appendix E.9). The ordering is stable - Prime $\times$ Low best covered in every window, a subprime cell worst - while the magnitude narrows from 79.69 points in the crisis to 24.94 in the rate-hike window, in step with the marginal inflation that lifts the worst-covered cells; crisis worst-group coverage is 10.31%. Along the score axis the surface is sharper than the predefined cells reveal (§4.2; Appendix E.5), and a depth-three segmentation of the pooled audit rows isolates a pocket holding 5.2% of them at zero coverage (Appendix E.8). A model predicting miscoverage from the calibrated score alone reaches an AUROC of 0.992 to 0.997 across held-out regimes, with the full specification adding at most 0.001 (Braun et al., 2025; Zhou et al., 2026) (Appendix E.6). Under SCP miscoverage is thresholded on the true-label nonconformity score. The calibrated probability alone nevertheless predicts it almost perfectly here. The headline 90% describes the over-covered low-risk centre, not the high-risk tail on which screening acts, and the window whose marginal coverage sits nearest nominal is the one whose subgroup and score-conditional coverage is worst.

These results cohere into the two-level taxonomy of §2.4. The first level is intrinsic: an imbalance-driven collapse fixed at the calibration split and deepened by recalibration, showing at once as class-conditional shortfall, subgroup heterogeneity and pocket-level failure, indifferent to distributional stability. The second is distributional: a regime-dependent modulation of the marginal and subgroup rates superimposed on it, running upward here. The answer to RQ1 is their interaction. Marginal coverage is preserved or inflated while class-conditional, subgroup and pocket-level coverage fail throughout and temporal coverage reverses sign for twenty-seven months during the crisis (§4.3). The two levels separate by sign and by dependence on time, so the failure decomposes into distinct kinds. The severest pockets belong to the intrinsic level, tracking score position and aligning with no single cell or regime, while the aggregate drift belongs to the distributional one. Neither is specific to the scorer or evaluation sample: an isotonic-calibrated logistic regression reproduces both, holding delinquent-class coverage between 0.14% and 1.43% and drifting from $+1.08$ to $+5.80$ percentage points, though it does not reproduce the narrowing of the subgroup CovGap, which stays between 67 and 77 points (Appendix E.11). Excluding code-01-affected loan-months moves marginal coverage by at most 0.99 points and leaves the collapse and the CovGap intact (Appendix E.3). Stating again the bound: the regime labels are a priori design hypotheses carrying the window qualifications of §3.2, so the ordering across four windows is descriptive rather than estimated.

4.2 What Each Conformal Variant Repairs (RQ2)

No variant evaluated here restores the coverage guarantee, because none restores exchangeability (§2.3). What separates them is which failure each design mechanism of §3.5 reaches, what it moves incidentally, and what it leaves behind.

Table 4.2: Method × regime reliability across the four test regimes

Method	Regime	MC (%)	Dev (pp)	$\widehat{\mathrm{Cov}}(y{=}1)$ (%)	Empty (%)	CovGap (pp)	WGC (%)
SCP	Crisis	90.51	$+0.51$	0.0007	8.52	79.69	10.31
	Normal	94.55	$+4.55$	0.0002	4.82	47.07	42.93
	COVID	94.83	$+4.83$	0.0000	4.29	31.97	58.03
	Rate hike	96.41	$+6.41$	0.0001	3.08	24.94	65.06
Mondrian (FICO × LTV)	Crisis	89.50	-0.50	0.98	9.34	5.26	84.74
	Normal	94.27	$+4.27$	0.79	5.01	7.34	87.11
	COVID	93.90	$+3.90$	0.18	5.18	5.04	90.35
	Rate hike	95.29	$+5.29$	0.20	4.17	6.15	92.29
APS	Crisis	88.99	-1.01	29.57	9.32	3.92	86.08
	Normal	89.46	-0.54	29.10	9.52	1.28	88.72
	COVID	89.14	-0.86	17.15	9.56	1.72	88.28
	Rate hike	89.58	-0.42	21.22	9.65	1.57	88.43
DtACI	Crisis	90.42	$+0.42$	0.0007	8.64	—	—
	Normal	93.81	$+3.81$	0.0002	5.58	—	—
	COVID	94.30	$+4.30$	0.0000	4.84	—	—
	Rate hike	95.98	$+5.98$	0.0001	3.52	—	—

Note. Test-window results; CovGap and WGC are taken over the ten FICO × LTV cells. CovGap and WGC were not evaluated for the adaptive method, which adjusts a single global threshold over time (—). ACI is reported in the text and in Appendix E.10.

The marginal overcoverage drift of §4.1 survives in all but one variant. Mondrian on the fine partition tracks SCP closely, from 89.50% in the crisis to 95.29% in the rate-hike window, and is flagged in the same three regimes (Table 4.2). ACI and DtACI are run in the monthly-update configuration of §3.5 on contemporaneous labels, a deployment adaptation of the published per-observation procedures rather than an implementation of them (Gibbs & Candès, 2021, 2024), so their coverage figures here are empirical diagnostics and not tests of a theorem. DtACI drifts as SCP does, from 90.42% to 95.98%, its adaptation showing in the target rather than the coverage: a final effective target of 0.887 against the nominal 0.900, at a long-run rate of 93.1%. Across the ACI grid that trade is monotone in the learning rate, long-run coverage approaching nominal, from 93.16% to 91.27%, only as the effective target falls from 0.871 to 0.781 (Appendix E.10). Only APS holds near 0.90 throughout, from 88.99% to 89.58% and never outside tolerance, its threshold not being a quantile of performing-class scores alone. Marginal coverage is, however, the dimension §4.1 showed to be least informative.

On the class-conditional dimension the methods separate sharply. SCP and DtACI leave delinquent-class coverage at or near zero in every regime and are indistinguishable at the reported precision: the adaptive threshold never moves far enough to reach the delinquent scores. Mondrian manages 0.18% to 0.98%: raising the threshold inside the high-risk cells admits a few delinquent loan-months but leaves class-conditional coverage at least two orders of magnitude below nominal. APS alone lifts it off the floor, to between 17.15% and 29.57% against its 37.60% calibration value, consistent with its cumulative-probability score avoiding the same majority-dominated true-label quantile (Romano, Sesia, & Candès, 2020). The repair is partial and regime-sensitive, between a sixth and three-tenths of delinquencies against a nominal 90%, and weakest for every method in the COVID window, where the forbearance regime can alter realised transitions relative to the signals the sets are built on (§3.2).

Two methods narrow the subgroup CovGap, by different routes. Where SCP's CovGap runs from 79.69 points in the crisis to 24.94 in the rate-hike window (§4.1), Mondrian compresses it to between 5.04 and 7.34 and APS to between 1.28 and 3.92 (Table 4.2). Outside the crisis Mondrian's CovGap is set by an over-covered cell, its worst-covered cell clearing nominal outright in the COVID and rate-hike windows. Mondrian equalises coverage across the predefined cells while leaving class-conditional coverage inside them near zero. APS narrows the CovGap from both ends as a by-product of closing the class-conditional shortfall, lifting worst-group coverage to between 86.1% and 88.7% while pulling the over-covered majority cell down, from SCP's 99.03% to 89.99% on the calibration window. Mondrian's equalisation is only as good as its partition, the coarser variants on FICO, LTV or census division leave crossed-grid CovGaps of 14 to 76 points (Appendix E.4).

Along the score axis the surface those summaries conceal comes into view (Figure 4.2). APS holds between 85.9% and 90.0% in every score decile, whereas SCP and Mondrian over-cover the three lowest deciles above 99.7% and collapse in the highest: crisis-window top-decile coverage is 14.2% under SCP and 49.2% under Mondrian against 85.9% under APS (Appendix E.5).

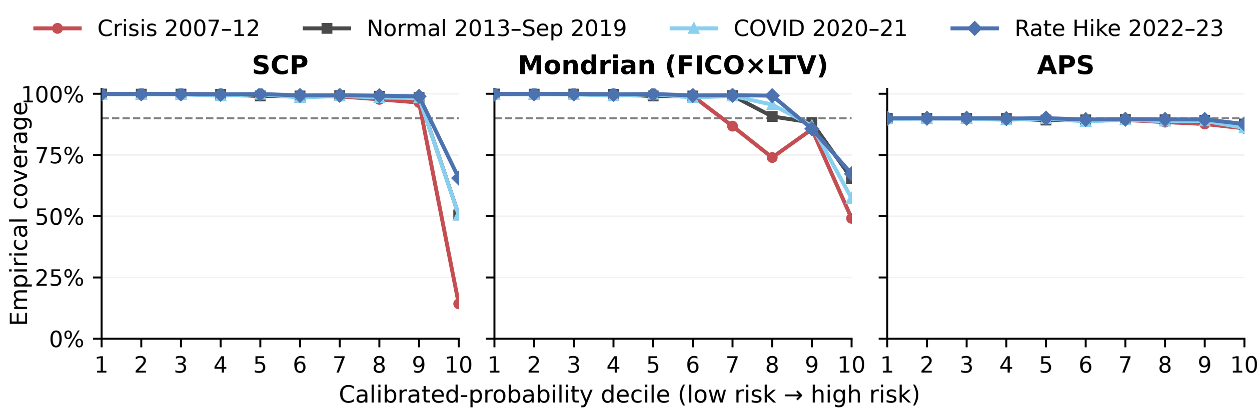

Figure 4.2: Empirical coverage across within-regime, equal-frequency calibrated-probability deciles, by method and regime; dashed line: 90% nominal.

The miscoverage model of §4.1 separates the static pair: under Mondrian the score alone averages 0.953, rising to 0.995 under the full specification, so its residual failure is cell-structured where SCP's is not, while under APS the model is near-random at 0.55, subject to the randomisation caveat of §3.6 (Appendix E.6). Concentration curves put that in audit terms: on the two regimes for which they were computed, the concentration area (3.7) runs from 0.955 to 0.977 under SCP and Mondrian against 0.533 to 0.579 under APS (Appendix E.7), so their failure is concentrated where a targeted audit would look. APS's residual failure is milder but not uniform: its worst depth-three pocket reaches 18.81% miscoverage, its per-cell miscoverage stays between 9.9% and 14.9% against SCP's 1% to 89%, and conditional on a calibrated probability of 5% or more it rises to between 14.3% and 17.5% - the high-risk region on which screening concentrates (Appendices E.9 and E.8). That near-uniformity is a practical approximation to conditional coverage, formally unattainable for any split conformal method (§2.2), not the property itself. And because the surviving pockets are defined by score position and cut across cell-regime combinations, Mondrian's cell-wise calibration leaves one at 99.98% miscoverage intact.

The costs show in the sets a pipeline would act on. SCP's positive-singleton rate rounds to zero in every window and Mondrian's never exceeds 0.04%, so neither can raise a flag in practice. APS is the only variant emitting delinquent-label-bearing sets at visible rates, on 0.53% to 1.54% of loan-months, which is what lets it cover the minority class. It pays for that with a non-committal rate of 10.16% to 10.75% under the randomised construction against SCP's 3.1% to 8.5% (Appendices E.1 and G.6). Mondrian's cost falls elsewhere: its cell-wise thresholds reorder loans against the global probability and capture fewer delinquencies at an equal non-auto-clear budget (§4.4).

The answer to RQ2 is a division of labour with spillover rather than a ranking. Each variant has one primary mechanism - APS the class-conditional collapse, Mondrian the subgroup heterogeneity that is its second face, ACI and DtACI the Level 2 drift, SCP none - and only APS reaches materially past its own, compressing the subgroup CovGap and flattening the decile profile while holding the marginal rate nearest to nominal of the four. None reaches every dimension, each leaves a residual failure of its own, and every repair is bought with abstention, selection efficiency or a lowered target. The attribution is robust to the scorer and sample restriction. An isotonic-calibrated logistic regression preserves it for the three static methods, with Mondrian's residual CovGap at 4.79 to 7.29 points and APS's class-conditional coverage at 15.5% to 29.0%, and code-01-restricted re-evaluation leaves the ordering unchanged (Appendices E.11 and E.3). As stated earlier: ACI and DtACI were not re-evaluated under the alternative scorer, so their attribution rests on the primary pipeline alone.

4.3 What a Standard Monitoring Process Sees (RQ3)

RQ3 asks what a deployed monitoring process, built around the population stability index (Siddiqi, 2017), would see of the failures of §4.1 and §4.2. The answer is almost nothing, for two separate reasons. The dominant component of conformal failure is not a distributional movement at all, so no statistic defined on the size of a movement can reach it. And the component that is a movement runs in the benign direction, the index ordering the four regimes in reverse of their conditional reliability.

The first reason is visible at the calibration anchor. On the calibration window the test distribution is the reference, so the index is identically zero. At that same split the delinquent class is covered at 0.0039% and the worst-covered cell already sits 82.58 percentage points below nominal (§4.1). No finer band recovers this: a statistic measuring the magnitude of distributional change cannot register a failure that is complete when the change is zero. The entire intrinsic level of the taxonomy therefore lies outside its domain, by definition of the statistic.

Across the four test regimes PSI does vary, and the second reason appears. It rises from 0.0062 in the crisis window to 0.1988 in the rate-hike window, and marginal coverage rises with it, so on its face the statistic behaves as intended. But the deviation it tracks is overcoverage, and against the conditional measures its ordering is exactly reversed. The window it rates most stable, the crisis, is the one with the widest subgroup CovGap at 79.69 points and the lowest worst-group coverage at 10.31%; the window it flags hardest, the rate-hike regime, has the narrowest CovGap and the highest worst-group coverage (Table 4.1; Appendix F.5). The reversal is exact across all four windows. The crisis's 65.8% rise in mean nonconformity score meets the lowest PSI of the four, the binned bulk barely redistributing while the tail's composition turns over (Appendix F.1). All four a priori regime labels of §3.2 are consistent with this evidence and the realised positive rate, the stable-drift label the most qualified (Appendix F.2).

The blindness extends from the cross-section to time, and the two regime onsets bracket it from opposite directions. Crisis-window aggregate coverage sits inside the ± 2 percentage-point band (§4.1) and its window-level PSI corroborates that stability. Yet the window's own positive rate climbs from 1.3% in January 2007 to 3.6% by December 2008, full twelve-month rolling coverage falls below nominal for twenty-seven months from January 2008 through March 2010, reaching a minimum of 89.0% in April 2009. And the single-month coverage bottoms at 88.7% in March 2009 - a sustained reversal of sign, inside the band in magnitude - before the later overcoverage absorbs it into the regime average (Figure 4.3). The 2020 onset is the complementary case: the monthly positive rate spikes above 5%, roughly double the crisis window's average of 2.53%, yet single-month coverage dips only to 91.7% in April 2020 and never crosses nominal, because forbearance decoupled the onset's realised transitions from credit deterioration (Conference of State Bank Supervisors, 2020; Federal Housing Finance Agency Office of Inspector General, 2020). Computed once per window over pooled observations, the index sees neither.

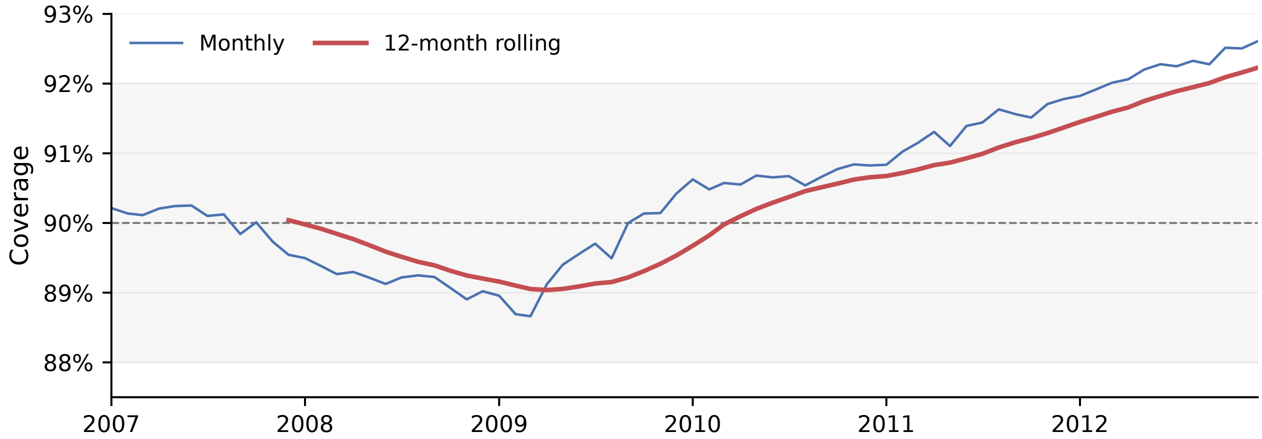

Figure 4.3: Monthly SCP coverage and full twelve-month rolling coverage in the crisis window; dashed line: 90% nominal; shading: ± 2 pp.

Each failure it misses maps onto a measure §3.6 already defines: class-conditional coverage for the intrinsic collapse, the subgroup CovGap and worst-group coverage for its heterogeneity, rolling temporal coverage for transient breaches. All three read the realised label, which under the twelve-month forward window is observed only once that window closes (§3.1), so they are lagging audits confirming a reliability state roughly a year after it begins rather than the real-time signal PSI is taken to be. The position at the close of RQ3 is that asymmetry: outcome-free drift signals can miss the least reliable regime, while the diagnostics faithful to the observed failures require realised outcomes and arrive in arrears. What monitoring design could follow is proposed in §5.3.

Two checks bound the result. The endpoint dissociation also appears without feeding the index the realised label (§3.6): the label-free feature-level variant ranks the crisis most stable on all four origination characteristics, so a feature tripwire places the least reliable window at its stable end, though it is not silent there, its note-rate index at 0.1364 crossing the investigation band (Appendix F.3). Nor is the pattern specific to the primary scorer: under an isotonic-calibrated logistic regression the crisis again carries the lowest PSI, at 0.0084, with the widest CovGap, and the highest-PSI regime, at 0.26, again the narrowest, while the two middle regimes swap on a 0.19-point margin (Appendix F.4). What replicates is the endpoint dissociation, not a monotone inverse relationship, which four observations could not establish in any case. One thing not tested here: a stability measure restricted to the threshold region or conditioned on class. Whether such a measure would realign with conformal coverage is a conjecture the evidence motivates.

4.4 Operational Translation and the Selection-Efficiency Null (RQ4)

The identity of §2.6 fixes what to expect: a clearance rule whose region is a lower interval of the score, cut at one threshold held fixed over the period compared, routes exactly as a threshold on that score does. The operational question is therefore not whether conformal routing screens well, since it inherits whatever the probability beneath it achieves, but whether it screens differently. The matched-budget comparison of §3.6 is the test.

It confirms the null and isolates one exception. At their natural operating points SCP and APS capture exactly the delinquencies selected by the probability ranking at the same non-auto-clear budget, the equal-budget gap standing at $+0.000$ percentage points in every regime (Table 4.3; Appendix G.4). That zero is not an empirical finding but the numerical signature of the identity - the two rules are the same rule - and because it holds at a matched budget it is invariant to any relative weighting of review cost against the cost of a missed delinquency. The exception is the informative result: Mondrian captures strictly fewer delinquencies than the baseline at the same budget, by 8.74 points in the crisis window and by 5.82, 3.73 and 5.43 thereafter, a conformal construction performing measurably worse than the model it is built on. The selection value of the layer is zero where its routing is monotone in the score and negative in the one construction where it is not.

Table 4.3: Operational routing summary by method and regime at each method's natural operating point

Method	Regime	Review (%)	Capture (%)	FAC (%)	$\widehat{\mathbb{P}}(Y{=}1 \mid \mathrm{AC})$ (%)	Rv. prec. (%)	Rv. lift	Δ (pp)
SCP	Crisis	8.52	62.05	37.95	1.05	18.45	7.29×	+0.00
	Normal	4.81	58.12	41.88	0.67	18.46	12.09×	+0.00
	COVID	4.30	47.69	52.31	0.91	18.49	11.09×	+0.00
	Rate hike	3.08	50.81	49.19	0.53	17.15	16.49×	+0.00
Mondrian (FICO × LTV)	Crisis	9.35	54.84	45.16	1.26	14.58	5.76×	−8.74
	Normal	5.00	52.76	47.24	0.76	15.87	10.39×	−5.82
	COVID	5.16	45.76	54.24	0.95	14.74	8.84×	−3.73
	Rate hike	4.16	48.66	51.34	0.56	12.13	11.66×	−5.43
APS	Crisis	2.56	43.30	56.70	1.47	42.77	16.89×	+0.00
	Normal	1.77	43.60	56.40	0.88	37.60	24.62×	+0.00
	COVID	1.51	30.96	69.04	1.17	34.26	20.55×	+0.00
	Rate hike	1.03	34.98	65.02	0.68	35.43	34.06×	+0.00

Note. AC = auto-clear. Δ is the equal-budget capture gap (3.8) against a calibrated-probability threshold; the matched budget is the non-auto-cleared share, equal to Review for SCP and APS and above it by at most 0.03 pp for Mondrian. ACI and DtACI are excluded (moving thresholds). Capture and Review are deterministic; the APS rates in Table 4.2 use the randomised coverage construction and are not directly comparable. Five-percent operational subsample (§3.1).

The deficit originates in the construction that narrowed Mondrian's subgroup CovGap. A separate threshold within each FICO × LTV cell makes its clearance region a union of cell-wise intervals rather than one global interval, and those cut-points run from 0.0033 in the prime, low-LTV cell to

0.4201 in the subprime, high-LTV cell, a spread of 128.7 times (Appendix D.1). A subprime loan with a delinquency probability near 0.4 is cleared while a prime loan near 0.004 is reviewed, and the budget follows the inversion. Mondrian allocates 6.1% of the prime, low-LTV cell, whose delinquency rate is 0.7%, against 3.1% of the subprime, high-LTV cell at 7.7%, where SCP's single threshold allocates the same two at 1.1% and 58.5% (Appendix G.1).

The three methods sit at different operating points. Reading them correctly prevents mistaking queue size for screening power. Where each clearance band ends sets the difference: APS, routed under the deterministic $U = 1$ reduction of §3.5, auto-clears every loan below a calibrated probability of 0.100, six times SCP's 0.0162, so it runs the smallest review queue, 1.03% to 2.56% of loan-months, at the highest review precision, 34% to 43%, with review lift reaching thirty-four times the portfolio rate. SCP reviews 3.1% to 8.5% at roughly half that precision (Table 4.3). The profile resembles a quality ordering and is not one. Because every clearance rule monotone in the probability traces the same capture-against-budget frontier within a regime, APS attains higher precision by reviewing fewer loans, not by ranking them better. Figure 4.4 makes this exact: SCP and APS lie on their regime's probability-ranking frontier, while Mondrian lies below it. One further difference sits off the frontier: APS's monthly review budget is the least variable in level, though not once normalised (Appendix G.2), which is a staffing convenience, not a screening advantage.

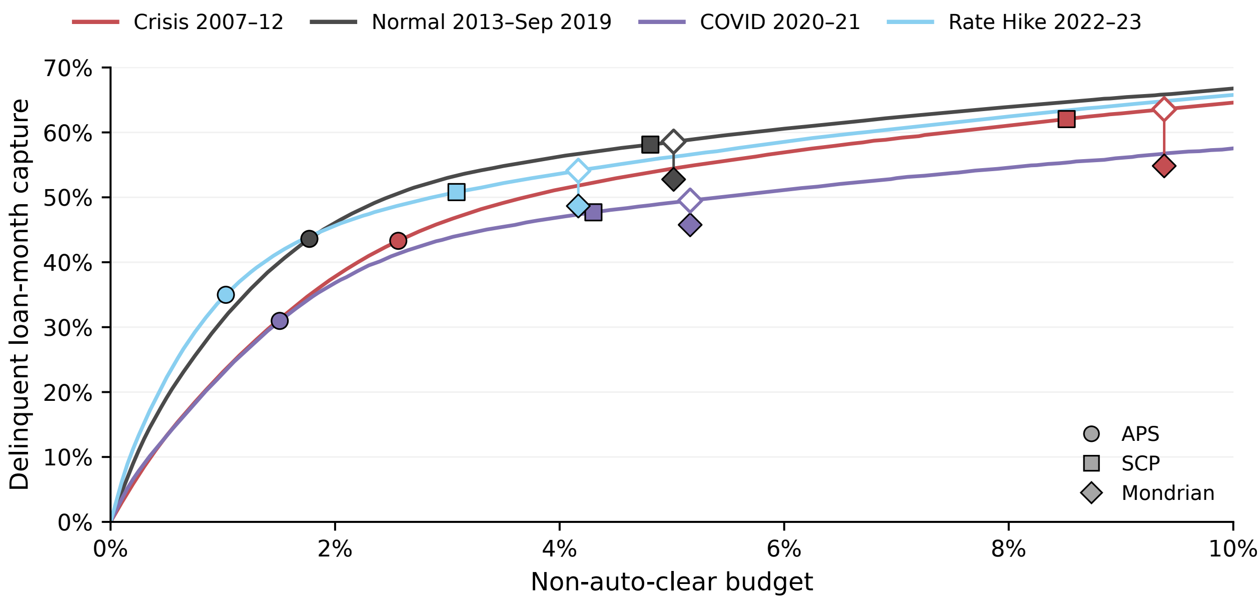

Figure 4.4: Probability-ranking capture curves and conformal operating points; hollow diamonds mark Mondrian's matched-budget baseline.

Under SCP and Mondrian, whose review set is the empty one, that capture is delivered by abstention. SCP covers the delinquent class at essentially zero yet routes 47.7% to 62.1% of eventual delinquents to review, because capture requires only that a loan was not auto-cleared where coverage requires the delinquent label to be in the set (§3.6). SCP's review queue is made of empty sets and APS's of the two-label sets of §3.5, non-committal in both cases and a positive flag in neither. Escalation peaks at 0.03% under Mondrian and 0.0003% under APS (Appendix G.6). The same composition fixes what the routing leaves behind. Between 38% and 69% of eventual delinquents are auto-cleared, 38% to 52% under SCP, 45% to 54% under Mondrian and 56% to 69% under APS, and the auto-cleared pool carries a delinquency rate of 0.5% to 1.5% (Table 4.3). So abstention screens, but it does not certify what it clears. APS shows the same separation less directly, its deterministic routing capturing 31% to 44% of delinquents against the 17% to 30% class-conditional coverage of its randomised sets (§4.2), which is a comparison across two constructions rather than a like-for-like gap, but one pointing the same way. The decision layer does not reproduce the class-conditional collapse; it partially circumvents it.

That fixes what the screening is without restoring a selection case. The capture SCP achieves through abstention is exactly what probability thresholding achieves at the same budget, so the screening is real but not attributable to the conformal layer, and the apparent advantages are level differences with no efficiency content: SCP captures roughly three million more crisis-era delinquencies than APS only by reviewing some thirty-eight million more loan-months (Appendix G.3). The answer to RQ4 is that relative to probability thresholding at an equal non-auto-clear budget the selection value of SCP and APS is exactly zero and Mondrian's is negative, so any case that remains for the conformal layer must rest on something other than selection efficiency (§5.2). The result is not an artefact of scorer or sample: under an isotonic-calibrated logistic regression the equal-budget gap stays exactly zero for SCP and APS, Mondrian's deficit stays negative in every regime though shallower, averaging 4.3 points against 5.9, and the non-auto-clear rates replicate within a percentage point throughout (Appendix G.5). The identity is invariant to the prepayment restriction by construction, depending only on the monotonicity of the clearance rule in the score.

5 Discussion

This discussion explores the consequences of the previously established failure patterns. Specifically, §5.1 analyses the results using the two-level taxonomy; §5.2 evaluates the contribution of CP once its selection value is nullified; §5.3 proposes the monitoring dissociation as a possible design; and §5.4 delineates the chapter's inferential limits. No new empirical measurements are introduced.

5.1 The Taxonomy as an Explanatory Lens

So far the two-level taxonomy has served to sort the observed failures into kinds (§2.4; §4.1). Its value is not only descriptive: once failure is decomposed into an intrinsic and a distributional level, the cross-question pattern of Chapter 4 becomes a connected set of consequences.

Three of those consequences follow from Level 1 alone. Because Level 1 is complete at zero distributional movement, a monitor defined on the magnitude of movement is blind to it by construction rather than by miscalibration, as §4.3 shows at a stability index of exactly zero. Because marginal coverage is the class-conditional mixture (2.5), in which the minority weight here is at most three percentage points, minority collapse cannot register marginally. And because clearance under a rule monotone in the calibrated score is a function of the same quantity whose geometry defines Level 1, the decision layer cannot reorder loans around the failure. That entailment binds the ordering, not the disposition: abstention routes roughly half to three-fifths of eventual delinquents to review while class-conditional coverage stays near zero (§4.4), so the operational stage inherits the ordering, not the collapse.

Which evaluated repair reaches which failure is, by contrast, an empirical mapping. Because Level 1 is a property of a threshold calibrated on a majority-dominated score mixture (He & Garcia, 2009; Löfström et al., 2015), a direct repair can act either on the pooled score geometry or on the pooling of calibration scores. Of those two routes only the first is evaluated here: the adaptive-prediction-set score lifts class-conditional coverage to between 17% and 30%, where cell-wise Mondrian calibration reaches only 0.18% to 0.98% and the adaptive procedures not at all (§4.2). This mitigation is observed on this panel, not guaranteed by the construction (Romano, Sesia, & Candès, 2020). The second, label-conditional calibration (Vovk, 2012), targets the collapse directly by estimating a separate threshold within each label. Leaving it unrun is a scope choice: the within-class scarcity that can make per-class thresholds coarse or inefficient (Ding et al., 2023) does not bind at this calibration size (§3.5). Two consequences can nonetheless be stated without running it: frozen per-class thresholds remain exposed to Level-2 changes in the within-class score distributions, whose direction and magnitude are not identified here. And the non-auto-clear region remains a single upper tail of the calibrated probability, so the matched-budget ordering identity of §2.6 still applies. What remains empirical is the out-of-period class coverage and the size and composition of that non-auto-clear region, referred to in §6.3.

Subgroup heterogeneity is the second face of Level 1, equalised by the Mondrian construction on risk-defined cells while the within-cell collapse stays essentially in place (§4.2). Under the split baseline the severest pockets track score position and align with no single segment or regime, the full specification adding at most 0.001 of discrimination beyond the score (§4.1). Under Mondrian the residual is cell-structured, the score alone identifying miscoverage at 0.953 against 0.995 under the full specification, and a pocket at 99.98% miscoverage survives (§4.2). Conditioning does not dissolve pocket-level failure, but it relocates it into the partition conditioned on, as a level generated by threshold geometry should once that geometry is re-cut.

The same lens places the adaptive-inference literature: procedures built to restore aggregate coverage under temporal shift (Gibbs & Candès, 2021, 2024) act on the distributional level by construction and are silent on the intrinsic one, their adaptation surfacing in the lowered effective target rather than in any recovery of class-conditional coverage (§4.2).

This reframes what the predictability of the results means. Neither level is itself new: the majority-class bias of split conformal prediction under imbalance is documented (Löfström et al., 2015), the class-conditional target it fails is well posed (Ding et al., 2023), and that coverage drifts once exchangeability is violated was never in doubt. The contribution is their joint decomposition, which converts a foreseeable qualitative fact into a map assigning each failure to a level, identifying which repair reaches it, and pricing that repair operationally. What generalises is the separation principle: under structured non-exchangeability, distinct reliability dimensions are repaired, if at all, by distinct mechanisms at distinct costs. For this panel, that is an empirical regularity; as a mechanism, it is a claim about regimes of this kind.

5.2 What Conformal Prediction Contributes When Selection Value Is Null

The operational analysis ruled selection efficiency out (§4.4). What remains, I argue, is semantic and governance-related. The redirection is forced by that section's strongest result: the equal-budget capture gap for the monotone methods is an exact zero, the signature of an identity rather than a measurement more data could move. A conformal layer whose clearance region is a fixed lower interval of the score is therefore not a better way to rank loans for review: it does nothing the calibrated probability does not already do. The qualification is not cosmetic: Mondrian's region is a union of cell-wise intervals, so the identity does not cover it, and it captures 3.7 to 8.7 points fewer delinquents at the same budget (§4.4). Where the identity holds, conformal prediction offers not a different rule but a different representation of the same one.

Decision theory is consistent with the null. Kiyani et al. (2025) distinguish calibrated forecasts for expectation-maximising decisions from suitably designed prediction sets for a Value-at-Risk objective. The matched-budget null here follows from the ordering identity, not from that result. Their framework instead locates prediction-set value in a risk-averse utility objective, which these coverage-calibrated sets do not target.

Three affordances survive the null. None is a capability the probability lacks. Each re-presents information it already carries, in a form built for audit and routing rather than selection. The first is that a prediction set expresses reliability in the units of the outcome: containment of the true label is a per-instance event (Shafer & Vovk, 2008) that can be tallied at any granularity rather than absorbed into an aggregate calibration statistic. The gain is realised only when that tally is read conditionally, since the marginal one is the aggregate §4.1 showed to mislead.

The second is that the set types support decision categories (Sadinle et al., 2019). The four possible sets are mapped here onto clearance, review and escalation (§2.6), and which of the two non-committal sets a method emits is a property of its construction (§3.5; §4.4). By the identity this partition coincides with a set of probability thresholds, so the vocabulary is not a new decision boundary, but its contribution is an explicit set-valued partition. Zwart (2026) makes that object the unit of deployment analysis: the region-by-class table is the audit summary from which commitment, deferral and error-exposure rates follow, and rules matched on coverage still differ across them. This sits alongside the identity: it fixes which loans a given budget selects, the profile what that budget buys. This is the difference §4.4 measures.

The third affordance, and the one doing the operational work, is abstention. Screening under the baseline runs through the non-committal disposition rather than through coverage of the delinquent class (§5.1). Declining to decide is not specific to conformal prediction - it is the reject option of selective prediction (Chow, 1957, 1970; El-Yaniv & Wiener, 2010; Geifman & El-Yaniv, 2017) - and by the same identity the non-committal region is again a probability band. Its contribution is formal: the construction emits a set type that the policy can tally as a non-decision rather than a discretionary carve-out. Routing it to review lets the decision layer partially circumvent the collapse it cannot repair, at non-auto-clear rates replicating under the logistic benchmark within a percentage point (§4.4).

Two readings from the literature bear on this role. The first is interpretive: where the set-valued classification framework treats the empty set as a nuisance to be designed away (Sadinle et al., 2019), here it is the operative disposition, carrying 9.41 of the baseline's 9.70 percentage points of calibration miscoverage (§4.1). The second reading is adverse. Jones et al. (2021) show that abstaining on low-confidence cases can widen accuracy gaps between groups and even lower accuracy on the group already worst served. Their result does not transfer directly to the monotone methods, for a structural reason: because that region is a band on the calibrated probability, review concentrates where predicted risk is highest, SCP allocating 58.5% of the subprime, high-LTV cell against 1.1% of the prime, low-LTV one. The warning transfers to the repair instead. Mondrian inverts that ordering, allocating 6.1% of the prime, low-LTV cell against 3.1% of the subprime, high-LTV one (§4.4): equalising the group reliability metric moves review capacity away from the segment carrying the risk. That disparity hazard attaches to the conditional repair, not to abstention as such.

A fourth candidate role - subgroup guarantees read as fairness constraints (Romano, Foygel Barber, et al., 2020) - I do not develop. The cells here are risk segments, not protected attributes, so what the pipeline equalises is subgroup reliability, and no protected-class criterion is tested. What survives is the separation Romano, Foygel Barber, et al. (2020) themselves insist on between risk assessment and the policy acting on it: conformal prediction makes per-segment reliability explicit and monitorable, a governance property even where it is not a fairness one.

Taken together, these arguments point to a potential role for conformal prediction as an audit and routing interface over an existing scorer, not as a better selector or a substitute for tuning that scorer's cut-off. The results temper even that: abstention screens without certifying what it clears, auto-clearing between 38% and 69% of eventual delinquents, and the evaluated variant covering the minority class best is the least reliable where screening acts, APS miscoverage rising from roughly a tenth overall to between 14.3% and 17.5% above a calibrated probability of 5% (§4.2; §4.4). Thus, the focus is on an honest way of expressing a decision to abstain, rather than on safety guarantees.

5.3 Implications for Model Risk Management

The monitoring analysis established a dissociation and stopped short of a prescription (§4.3). Here I propose one. The hazard is concrete: a credit-risk function adding conformal prediction to an existing pipeline could inherit a stability monitor read against conventional bands (Siddiqi, 2017) and take its quiet signal as reassurance about a reliability it never measures. The decisive half of the dissociation needs no regime comparison: on the calibration window measured instability is exactly zero while the delinquent class is covered at 0.0039% and the worst cell sits 82.58 points below nominal (§4.1). The four-regime evidence adds a weaker second finding, the index also ordering the regimes against their reliability, with the endpoints surviving a change of scorer (§4.3). The incumbent statistic is therefore not merely silent about the dominant failure. On this evidence it can point the wrong way about the rest.

The proposal has two elements, one for each half. First, the global stability index warrants supplementing rather than retuning, by a drift signal localised to the decision threshold and stratified by predicted-risk band. The justification is not that such a signal would detect the intrinsic collapse - none can (§5.1) - but that the drift a label-free monitor can see is diluted over the score bulk below the threshold, which fills nine of the index's ten bins (§2.5). Restricting the signal to the threshold region, where clearance decisions concentrate and miscoverage is essentially a deterministic function of score position (§4.1), removes that dilution. The pipeline already emits one. Because the baseline returns only the performing singleton and the empty set, its miscoverage is exactly the abstention rate plus the positive rate times the share of delinquencies auto-cleared: 8.52 and 0.96 percentage points in the crisis window, 3.08 and 0.51 in the rate-hike one. The first term is the deployment score's mass above the threshold and needs no outcome; the second waits a full label horizon. That split motivates a two-speed design, read off the conformal layer's own output.

Second, because the only signals faithful to the intrinsic and subgroup failures are functions of the realised outcome, that tripwire has to be paired with lagged coverage audits - class-conditional, subgroup and rolling temporal - with the label latency carried explicitly (§4.3). The lag is a known cost of the slow leg. Supervisory guidance supports the logic of both elements without prescribing this two-speed design: outcomes analysis compares model outputs with realised outcomes, while ongoing monitoring assesses performance as conditions change, with timing and frequency tailored to model purpose, data availability and materiality (Board of Governors of the Federal Reserve System et al., 2026).

This recasts the conclusion of §4.2 as a proposal: auditing conformal reliability under shift calls for a panel of conditional diagnostics rather than a headline number, because the dimensions that fail are repaired, where at all, by different mechanisms at different costs (§5.1). But this proposal is stated, not validated: it rests on the mechanism of the dissociation, on four regime observations, and on the endpoint replication under one alternative scorer (§4.3). And the fast signal it rests on is reported here but never tested for lead time.

5.4 Bounds on This Chapter's Claims

For emphasis, three bounds delimit this chapter's claims, distinct from the data-level qualifications bounding Chapter 4. First, the empirical half of §5.1 - which repair reaches which failure - rests on a single panel, so its status as a general principle is argued, not validated. Second, the case in §5.2 is argued rather than measured: what conformal prediction affords where selection value is null is not a quantity this thesis estimates. Third, again, the design in §5.3 is motivated by the dissociation, not tested against it.

6 Conclusion

6.1 Answers to the Research Questions

RQ1. The failure decomposes into two kinds: an intrinsic level fixed at the calibration anchor by the pooled score geometry, and a distributional level superimposed by calibration-to-test movement, which here moves the aggregate upward. They separate by sign and by dependence on time, and their interaction inverts the headline: the regime whose marginal coverage sits nearest nominal is the least reliable by subgroup and by score position.

RQ2. No variant restores the guarantee, because none restores exchangeability. Each targets one mechanism and only one reaches past it: the adaptive prediction-set score reaches the class-conditional collapse and narrows the subgroup CovGap with it, cell-wise Mondrian calibration reaches the subgroup dimension alone, and the online-adaptive procedures reach the class-conditional collapse not at all, lowering their working level without bringing realised coverage back to nominal. None covers all five dimensions, and every repair is bought with abstention, selection efficiency, or a lowered target.

RQ3. It sees almost nothing. The intrinsic failure is already complete where measured instability is zero, so no statistic defined on the size of a movement reaches it. And across the test regimes the index ordered them in reverse of their reliability. The diagnostics that directly measure these failures need the realised outcome and arrive a full label horizon late, which motivates pairing a fast outcome-free tripwire with lagged coverage audits.

RQ4. It does not. At a matched review budget every clearance rule that is a fixed lower interval of the score selects what a probability threshold selects; the one that is not performs strictly worse. Screening does occur, but it is inherited from the probability beneath and runs through withheld decisions rather than positive flags. Whatever case remains is semantic and governance-related, not selection-efficient.

6.2 Contributions and Their Epistemic Status

I make five contributions that differ in terms of their epistemic status.

The first gives the thesis its title. That imbalance degrades class-conditional coverage is documented, and that coverage drifts once exchangeability is violated was, again, never in doubt. What is added is the joint pattern and its anchor. Marginal, class-conditional, subgroup, pocket-level and temporal coverage were measured on the same rules and panel. Under the split baseline the aggregate was met or exceeded everywhere while three conditional measures failed throughout and the fourth, temporal, reversed sign during the crisis. The failure is complete on the calibration window itself, on the rows the threshold is drawn from. Its conceptual half is the two-level taxonomy, separating what the calibration score geometry fixes in advance from what the regimes modulate, and so saying which repair reaches which failure. It is argued that this could serve as an organisational tool, though this is not demonstrated.

The second aspect concerns the perception of a deployed monitor, as its two halves differ in their informational value. The stability index is zero on the calibration window by construction while the intrinsic failure there is already complete: a statistic measuring the size of a movement cannot register a failure present at zero movement. That half is definitional. The other is empirical and weaker, as four observations cannot estimate a relationship, and only the endpoints of that ordering survive a change of scorer. As noted, two things rule out an artefact of how the index was fed: it read the conformity score at the realised label, which a live monitor would not have, and a label-free index on origination characteristics ranks the crisis most stable too.

The third is operational; its halves have different standing. That a clearance rule forming a fixed lower interval of the score selects, at a matched budget, what a probability threshold selects is an identity, not a discovery. The arithmetic is elementary and the difference exactly zero in every regime and scorer. Its value is to foreclose a natural expectation: the question becomes whether the layer screens differently, not better. The exception is measured: calibrating separately within borrower-risk cells breaks that monotonicity, and the rule then captures 3.7 to 8.7 points fewer delinquents at the same budget, a conformal construction performing worse than its own base model.

The fourth comes from reading the sets as dispositions. The split baseline covers the delinquent class at essentially zero yet routes roughly half to three-fifths of eventual delinquents to review: capture needs only that a loan was not cleared, coverage that the delinquent label be in the set. Abstention itself is not new and what the evidence adds is that in that baseline all the operational work runs through it and none through minority-class coverage - and that it screens without certifying, 38% to 69% of eventual delinquents being cleared automatically.

The fifth is a proposal, not a finding: if each dimension is repaired, where at all, by a different mechanism at a different cost, no single coverage figure certifies such a system. I establish the premise here but the standard is motivated, not validated.

All five describe one government-sponsored enterprise's conforming, fixed-rate book under a deliberately violated exchangeability assumption: they show how conformal reliability fails under stress, not how it behaves where it holds.

6.3 Future Research

Three programmes could follow, the first two on material already assembled.

The first is earlier detection. The discussion identifies the fast leg but does not test it: a stability statistic on the threshold neighbourhood rather than the score bulk, stratified by predicted-risk band. The test is lead time: whether it turns before the coverage it anticipates, early enough to act, rather than whether it agrees with the global index. The confirmatory leg is equally open: the audit assumed observed outcomes, which live deployment lacks, leaving unresolved what surrogate sustains it while outcomes mature. Foygel Barber et al. (2023) pose the same question in general form: whether a deployment's departure from exchangeability can be judged adaptively, before coverage is lost.

The second is joint repair and its price. Label-conditional conformal prediction, a Mondrian calibration by the outcome class, attains per-class validity directly (Vovk, 2012) rather than through the score. Running it against adaptive prediction sets would settle whether the direct route beats the score-based one, and at what price. The covariate side has generalisations: exact finite-sample coverage over overlapping groups, where the Mondrian partition used here must be disjoint (Gibbs et al., 2025), and a threshold function in place of the single moving threshold, which buys localisation with slower convergence (Zecchin & Simeone, 2024). Neither conditions on the realised class, so whether the two compose, and at what joint price, is open. Two implementation questions belong here: whether the monthly cadence is a free parameter or a confound under rare events given that a live system has only a year-old signal, and whether recalibration deepens the collapse causally, which only a controlled calibration-conformal comparison on the primary pipeline could establish.

The third is to make non-exchangeability empirical. Under joint covariate and conditional shift, the covariate-ratio correction of Tibshirani et al. (2019) is insufficient because it assumes invariant $P(Y \mid X)$. Unlabelled deployment covariates do not identify the conditional change. How much a delayed label stream would identify is unresolved. The coverage-gap bound is harder: its tractable route runs through pairwise distances between calibration and deployment observations and requires independence (Foygel Barber et al., 2023), which the repeated loan-months and overlapping labels here deny. Whether a surrogate survives that dependence, and whether the bound would have anticipated the crisis-onset shortfall observed here, is untested.

Marginal conformal validity is not a sufficient description of reliability in an imbalanced, shifting screening environment. As noted, the constructions available redistribute failures rather than remove them, each buying one dimension with abstention, selection efficiency, or a lowered target. Deployment therefore warrants conditional diagnostics and an account of what each repair costs.

A Theoretical Supplement

A.1 Finite-Sample Validity of Split Conformal Prediction

The guarantee (2.2) rests on one fact about ranks. Write $V_1, \ldots, V_n$ for the calibration scores and V_{n+1} for the test score. If these $n+1$ variables are exchangeable then, for any $\beta \in (0, 1)$,

$$\mathbb{P}\Big(V_{n+1} \leq \text{Quantile}\big(\beta;\, V_1, \ldots, V_n, \infty\big)\Big) \geq \beta, \tag{A.1}$$

and if ties occur with probability zero the same probability is at most $\beta + 1/(n+1)$ (Tibshirani et al., 2019, Lemma 1). Setting $\beta = 1 - \alpha$ and reading the inclusion rule (2.1) as the event in (A.1) gives (2.2) directly. For split conformal, Lei et al. (2018) give the analogous regression result; under their even split, the upper correction $2/(n + 2)$ equals $1/(n_{\text{cal}} + 1)$.

One condition attaches to each bound. The lower bound requires exchangeability of those $n+1$ scores and nothing more: Proposition 1 of Vovk (2012) asks nothing of the proper training set, which is held fixed, and nothing of the conformity measure beyond measurability. The upper-bound result additionally assumes zero tie probability; (Lei et al., 2018) impose a continuous joint residual distribution as a sufficient condition, while randomised tie-breaking recovers the rank upper bound when ties are present (Tibshirani et al., 2019, Rem. 2).

That condition fails here. The isotonic map of §3.4 is constant on each pooled block (Appendix C.2), so the split-conformal score carries atoms at the block levels. At the $n = 161{,}421{,}966$ conformal calibration rows and $\alpha = 0.10$ the quantile index $\lceil (n+1)(1-\alpha) \rceil = 145{,}279{,}771$ sets the threshold at a level 9.9×10^{-9} above $1 - \alpha$, and the band in (2.2) is $1/(n + 1) = 6.2 \times 10^{-9}$ wide. These both corrections are immaterial here.

The atom at the threshold is not immaterial. A share 0.005478 of the calibration scores equals the threshold exactly, so the empirical distribution function jumps from 0.897549 just below it to 0.903027 at it, straddling the required level; the rule $s \leq \hat{q}$ admits the atom whole. Realised coverage is therefore 0.903027, the standing offset of 0.30 percentage points read in §3.5 and §4.1. The APS score (3.3) carries a uniform random draw (Romano, Sesia, & Candès, 2020): at its selected threshold, no tie mass is visible and coverage attains 0.900000 (Table A.1).

Table A.1: Score granularity and realised coverage on the conformal calibration partition

Nonconformity score	Quantile level	Tie mass at $\hat{q}$	Realised coverage
Split conformal, $s = 1 - \hat{p}(y \mid x)$	0.900000010	0.005478	0.903027
Adaptive prediction sets (randomised)	0.900000010	0.000000	0.900000

Note. Conformal calibration rows of §3.4, $\alpha = 0.10$. Quantile level $\lceil (n + 1)(1 - \alpha) \rceil / n$; tie mass the share of scores equal to the threshold, realised coverage the share at or below it. Thresholds in Appendix D.1.

B Data, Panel, and Sample Construction

B.1 Data Release, Product Scope, and Panel Construction

The panel is built from Release 46 of the Standard SFLLD, issued 28 January 2026, whose origination and performance cut-off dates both fall on 30 September 2025 and which spans 107 origination quarters (Freddie Mac, 2026c). Ingested record counts reconcile to the release's published totals within 0.3%.

The Standard release holds fully amortising 10-, 15-, 20-, 30- and 40-year fixed-rate mortgages underwritten with verified or waived documentation, together with Relief Refinance loans and Home Possible loans funded from 1 March 2015. Adjustable-rate, interest-only, balloon and step-rate products, government-insured and reduced-documentation lending, deliveries under alternate agreements and loans carrying credit enhancement beyond primary mortgage insurance go to the companion Non-Standard release, last refreshed in January 2022 and not used here. The split follows credit-risk-transfer eligibility, so the exclusion is systematic rather than random (Freddie Mac, 2026b, 2026c). Two further qualifications attach. Loans between the baseline limit and the high-cost-area ceiling do appear and are flagged as super conforming (Federal Housing Finance Agency, 2026). And the Standard release is a living one whose inclusion criteria were last restated at Release 43, so the panel is a snapshot of one release rather than of a fixed population.

Table B.1 traces the construction. Origination features join on the loan sequence number, so every performance record has an origination match. The base population exceeds the analysis panel by 16,446 loans (0.04%), whose eligible months fall entirely inside the buffer periods at the two regime boundaries of §3.2; this final exclusion removes 3.60% of assembled rows. Every remaining loan-month then falls in exactly one of the six windows of Table 3.1, and no (loan, month) pair occurs twice.

Table B.1: From the Release 46 source files to the analysis panel

Stage	Rule applied	Loan-months	Distinct loans
Origination files	Release 46 Standard, 1999Q1–2025Q3	—	48,597,637
Performance files	all monthly records to 2025Q3	2,800,558,227	48,597,636
Base population	no zero-balance code; loan age ≥ 1; status current or 30–59 DPD; month in 1999–2023	2,408,959,003	46,664,954
Buffer removal	Oct 2019–Feb 2020; Oct–Dec 2021	−86,758,394	−16,446
Analysis panel	six chronological windows (Table 3.1)	2,322,200,609	46,648,508

Note. Loan-months does not apply to the origination file, which holds one record per loan (dash). The inner join attaching origination features is lossless, so it is not shown as a separate stage. Buffer removal is shown as a signed reduction; all other entries are running levels.

B.2 Label Construction and Measurement Diagnostics

Both events of (3.1) come from one disclosure column, Current Loan Delinquency Status: a categorical value corresponding to delinquency-day ranges, with `0` denoting current or less than 30 days delinquent, `1` denoting 30–59 days, `2` denoting 60–89 days, `3` denoting 90–119 days, and so on; `RA` denotes REO acquisition, with `R` used before Release 29 (Freddie Mac, 2026b, 2026c). Events are read only from rows with no zero-balance code, the activity condition the base population itself imposes, so a terminal record never supplies one - the source of the fast-default gap below. The status is computed by the Mortgage Bankers Association method from the due date of the last paid instalment and reported on a one-month lag, which applies to the base-population filter and to the label alike and so shifts the observation clock rather than the twelve-month distance between them.

Each positive event at month m is then propagated backwards to observation months $m - 1, \ldots, m - 12$: the operational form of the window in (3.1), and the source of the overlapping-label dependence of §2.3. Voluntary payoff and maturity carry zero-balance code 01 and close the record with no event, the competing-risk coding of §3.1 (Deng et al., 2000; Fine & Gray, 1999). Table B.2 derives the three measurement figures the main text asserts.

Table B.2: Derivation of the label and window measurement diagnostics

Claim	Construction	Value
Fast defaults (§3.1)	Loans terminating under zero-balance code 02 (third-party sale) or 03 (short sale or charge-off) with no active 60+ DPD or RA month on record, against loans carrying at least one positive label	Full ingested performance history: 4,377 of 240,233 ZBC-02/03 loans (1.82%); equivalent to 0.17% of distinct loans in the positive-label lookup.
Calibration boundary (§3.2)	Positive rate on calibration rows whose label window reaches into 2007 (observation month from January 2006) against rows whose window closes within 2006, the latter read as the uncontaminated counterfactual	51.8% of the window; 1.0904% against 1.1258%, so 1.1075% against a 1.1258% counterfactual: -0.0183 pp
Normal-window contamination (§3.2)	Share of the normal window whose label window reaches the COVID regime (observation month from March 2019), and the positive rate on that share against the remainder	9.31% of rows; 3.2211% against 1.3423%, a gap of $+1.8788$ pp

Note. The calibration counterfactual assigns the clean portion's positive rate to the later portion of the calibration window. It therefore quantifies the latter's contribution to the aggregate positive rate, not its effect on the conformal threshold. The threshold depends on the distribution of nonconformity scores, which is determined jointly by model scores and realised labels.

B.3 Sample Allocation and Deterministic Hashing

Every principal sample in Table B.3 is defined by a modular hash rule on the loan sequence number, giving loan-level deterministic membership. This is the mechanism behind the loan-level determinism of §3.1. Three hash functions are in use: DuckDB's built-in hash for the feature-drift sample, Polars' seeded hash for the model-fitting samples, and zlib's `crc32` from the score files onward. The independently hashed sampling schemes are not generally nested; CRC32-based samples may be nested or identical by construction. Only the `crc32` rules are reproducible independently of library version. One separation is load-bearing and it is exact: within the calibration window the residue class $\texttt{crc32} \equiv 0 \pmod{10}$ fits the isotonic calibrator and the nine complementary classes carry the conformal threshold, so no row does both jobs (§3.4, §3.5): Table B.3 sets out the principal samples, smaller single-purpose draws being documented where they are used.

Table B.3: Allocation of rows across analytical samples

Sample	Selection rule	Rows	Window and use
Full panel	—	2,322,200,609	Underlying analytical panel before task-specific windowing and sampling
iso_fit	$\texttt{crc32} \equiv 0 \pmod{10}$	17,922,155	Calibration: isotonic calibrator
cp_cal	$\texttt{crc32} \not\equiv 0 \pmod{10}$	161,421,966	Calibration: conformal threshold
2% discrimination	seeded hash 99, 20/1,000	41,518,218	Calibration and four test: AUROC and average precision (§3.4)
2% score drift	$\texttt{crc32} \equiv 3 \pmod{50}$	3,572,364	Calibration cp_cal rows: score PSI reference (§3.6)
5% audit	$\texttt{crc32} \bmod 1{,}000 < 50$	94,869,688	Four test: local reliability audit and operational routing (§3.6)
5% benchmark	$\texttt{crc32} \bmod 1{,}000 < 50$	12,385,067	Training split: logistic benchmark fitting and 2004 validation (§3.4)
10% feature drift	built-in hash $\equiv 0 \pmod{10}$	—	Calibration and four test: feature-level PSI (§3.6)
3% tuning	seeded hash 42, 30/1,000	7,450,958	Training: hyperparameter search (§3.4)
1% reliability	seeded hash 1, 10/1,000	1,797,233	Calibration: raw-score reliability tables and ECE (§3.4)

Note. iso_fit and cp_cal partition the calibration window exactly, 179,344,121 rows. The two five-percent samples share a rule: the benchmark's four test windows are the audit rows, and a further 8,972,803 calibration-window rows drawn the same way carry its calibrator and conformal partition (Appendix C.3). The score-drift row is the PSI reference alone; its matched test-window samples are sized in Table D.3. The ten-percent row count is not stored.

B.4 Feature Dictionary, Exclusions, and Leakage Screen

Several disclosure sentinels are not pure non-disclosure. The LTV and CLTV valid ranges depend on the origination regime, so a ratio valid for one vintage is out of range for another and is nulled by the rule that applies to its own. And the debt-to-income sentinel 999 conflates a ratio above 65%, an unavailable value and the blanket masking of relief-refinance loans, which is why the missingness flag is split by relief-refinance status (Freddie Mac, 2026a). The relief-refinance branch of that split is a thirtieth eligible feature, constant on the training window and dropped before fitting, leaving the twenty-nine features of Table B.4.

Table B.4: Feature dictionary

Feature	Definition	Coding and missing treatment
Static origination characteristics (16)		
`credit_score`	Disclosed Credit Score at acquisition	300–850 retained; 9999 and out of range to null
`original_ltv`	Original loan-to-value ratio	6–105 pre-2018Q2 non-relief, else 1–998; 999 and out of range to null
`original_cltv`	Original combined loan-to-value ratio, including disclosed secondary financing	As `original_ltv`, upper bound 200 pre-2018Q2
`original_dti`	Original debt-to-income ratio	$(0, 65]$ retained; 999 to null
`original_interest_rate`	Note rate at origination	Positive values retained
`original_upb`	Unpaid balance on the note date	Rounded to $1,000 at source
`original_loan_term`	Scheduled payment count from first payment to maturity	Positive values retained
`vintage_year`	Origination year, parsed from the loan sequence number	Categorical, 1999–2023
`vintage_quarter`	Origination quarter, same source	Categorical, 1–4
`census_division`	Census division of the property state (U.S. Census Bureau, 2010)	Nine divisions; territories assigned to the nearest, else Unknown
`loan_purpose`	Purchase, cash-out refinance, no-cash-out refinance, refinance not specified	Categorical; 9 and blank to Unknown
`occupancy_status`	Primary residence, second home, investment property	Categorical; 9 to Unknown
`property_type`	Single-family, condominium, PUD, co-op, manufactured housing	Categorical; 99 to Unknown
`number_of_units`	One- to four-unit property	1–4 retained, else null
`is_multi_borrower`	Binary transform of the borrower-count disclosure	1 if raw count > 1 (including $99 =$ Not Available); 0 otherwise
`first_time_homebuyer_flag`	First-time homebuyer indicator	Y / N / Unknown / Not applicable, the last where the status cannot apply
Dynamic current state at t (5)		
`current_upb`	Unpaid balance at t, interest-bearing plus deferred	Missing coerced to zero
`upb_rel_change_3m`	$(\mathrm{UPB}_t - \mathrm{UPB}_{t-3})/\mathrm{UPB}_{t-3}$	Null unless loan age ≥ 10 and unmodified, excluding the origination rounding window and post-modification balance jumps
`current_interest_rate`	Note rate in force at t, reflecting any rate modification	—
`rate_spread`	Current less original note rate	Null if either rate is null
`is_30dpd_at_t`	Reported status is 30–59 days delinquent	Binary

Table B.4, continued

Feature	Definition	Coding and missing treatment
Twelve-month delinquency history (3)		
`n_times_30dpd_last_12m`	Count of 30-day-delinquent months in $[t-12,\ t-1]$	0 where no event
`max_dlq_last_12m`	Maximum encoded status over the same window	RA and legacy R coded 99; 0 where no event
`months_since_last_dlq`	Months since the most recent delinquency in the window	1–12; 999 where no event
Disclosure missingness flags (5)		
`fico_missing`	Credit score cleaned to null	Binary
`ltv_missing`	Original LTV cleaned to null	Binary
`cltv_missing`	Original CLTV cleaned to null	Binary
`dti_missing`	Original DTI cleaned to null	Binary
`dti_missing_non_relief_refi`	DTI null on a non-relief-refinance loan	Binary; isolates the informative branch from relief-refinance masking

Note. Credit-score/LTV/CLTV/DTI sentinels are cleaned to null and never imputed (Little & Rubin, 2019; Rubin, 1976).

A training-window Pearson screen of selected numeric panel fields, flagging any $|r| > 0.30$, returns the current and lagged 30-day-delinquency indicators alone, at $r = 0.330$ and 0.326; both are observable at t and are retained. The screen is a first-pass diagnostic, not the exclusion criterion (Kuhn & Johnson, 2013): the modification flag correlates at only $r = 0.076$ yet is excluded, the two grounds turning on what a field records and when it is collected rather than on how strongly it correlates. Table B.5 lists the eight withheld fields, four on each ground.

Table B.5: Fields withheld from the feature set

Field	Ground	Measurement
`modification_flag`	Servicing	Set in the month a modification settles and in every month after
`payment_deferral_flag`	Servicing	As above, for the deferral programme introduced in 2020
`deferred_upb_ratio`	Servicing	Non-interest-bearing share of the balance; non-zero only after a deferral or modification
`in_workout_plan_flag`	Servicing	Forbearance, repayment or trial-period enrolment; disclosed from January 2014
`loan_age`	Temporal	Scheduled-payment age; reset after modification
`remaining_months_to_legal_maturity`	Temporal	Months to legal maturity; modified maturity after modification
`amortization_ratio`	Temporal	Current over original balance; strongly seasoning- and schedule-related
`channel`	Temporal	Granular broker/correspondent reporting begins in 2008 (Freddie Mac, 2026a)

B.5 Borrower-Risk Cells: Sizes and Event Rates

Table B.6 derives the cell sizes and event rates §3.3 quotes, on the calibration window against which every later window's subgroup coverage is read (§3.2). Loan-month counts use the full window; Appendix D.1 reports the same cells on the nine-tenths conformal partition used for the Mondrian thresholds, with correspondingly smaller counts.

Table B.6: Borrower-risk cells on the calibration window

Cell	Distinct loans	Loan-months	Share (%)	Positive rate (%)
Prime × Low	5,298,716	100,930,335	56.28	0.25
Prime × Moderate	476,838	8,695,106	4.85	0.82
Prime × High	29,042	469,140	0.26	1.68
Near-Prime × Low	2,645,334	47,166,131	26.30	1.42
Near-Prime × Moderate	542,602	9,515,042	5.31	3.13
Near-Prime × High	37,843	556,319	0.31	4.27
Subprime × Low	526,427	8,848,556	4.93	5.01
Subprime × Moderate	145,606	2,442,791	1.36	8.27
Subprime × High	8,587	103,181	0.06	9.29
Sentinel	—	617,520	0.34	—

Note. Cut-offs are as §3.3; shares are of the calibration window. The sentinel cell is a residual: its loan-month count is the window's 179,344,121 rows less the nine crossed cells, and its distinct loans and event rate are not separately recorded. Every crossed cell clears the 500-loan viability floor checked at panel construction, the smallest by a factor of seventeen.

C Model Est., Probability Calibration, and Benchmark

C.1 LightGBM Configuration, Tuning, and Feature Attribution

The model is fitted on the 166,337,801 loan-months of 1999–2003, with the 81,753,843 rows of 2004 held back for early stopping (§3.2). On the fitting rows the negative-to-positive ratio is 164,298,584 against 2,039,217, which fixes the `scale_pos_weight` of Table C.1; the full training window of Table 3.1 implies 85.51, the gap being the 2004 rows and their positive rate of 1.01% against 1.23% over 1999–2003. Hyperparameters are searched by tree-structured Parzen estimation (Akiba et al., 2019; Bergstra et al., 2011) over eighty trials on a three-percent loan-level sample, each trial scored by validation AUROC on that sample's 2004 rows with a 600-round cap, 40-round early stopping, and median pruning. Two model hyperparameter values differ in the full-scale refit: `min_child_samples` is multiplied by the integer scale factor 33, obtained as `int(166,337,801 / 4,998,381)`, approximately preserving the sampled minimum-leaf-size fraction at full scale; and `max_bin` is raised from 127 to 255.

Table C.1: LightGBM specification as fitted

Setting	Value	Source
Objective / metric	binary log-loss / AUROC	fixed
scale_pos_weight	80.5694	fitting rows
learning_rate	0.05364	searched, $[0.045, 0.09]$ log
num_leaves	128	searched, $[63, 255]$ log
max_depth	9	searched, $[6, 9]$
feature_fraction	0.7688	searched, $[0.65, 0.95]$
bagging_fraction	0.6560	searched, $[0.65, 0.95]$
bagging_freq	1	searched, $[1, 3]$
min_gain_to_split	0.5227	searched, $[0, 1]$
lambda_l1	0.007502	searched, $[10^{-4}, 5]$ log
lambda_l2	0.8507	searched, $[10^{-4}, 25]$ log
min_child_samples	68,607	2,079 searched $\times$ 33
max_bin	255	127 in search
Boosting rounds	646 of 1,000 (early stopping, 25 rounds)	validation AUROC 0.92711
Seed	42 (search and fit)	fixed
Categorical handling	5 label-encoded string, 3 integer	design (§3.3)

Note. For num_leaves, the search upper bound was additionally constrained to $2^{\texttt{max\_depth}}$.

Table C.2 ranks the ten features carrying the largest mean absolute SHAP attribution (Lundberg & Lee, 2017; Lundberg et al., 2019). The ranking is stable across independent draws, at a Spearman correlation of 0.9995 between two seeds with the same ten features in both top tens. Two aspects of it bear on choices made elsewhere. Credit score and original LTV rank first and fourth overall, and first and third among origination characteristics, so the borrower-risk partition of §3.3 crosses two of the three origination characteristics with the largest mean absolute SHAP attributions. And four of the twenty-nine features receive no attribution at all, the booster never splitting on them: rate_spread, non-zero on 0.02% of training rows because a fixed-rate note carries no spread absent rate-modification activity, and three of the five disclosure missingness flags, the other two ranking twenty-fourth and twenty-fifth.

Table C.2: Leading SHAP feature attributions, calibration window

	Feature	Mean |SHAP|		Feature	Mean |SHAP|
1	credit_score	0.7665	6	months_since_last_dlq	0.1896
2	is_multi_borrower	0.3986	7	original_dti	0.1886
3	current_interest_rate	0.3798	8	original_upb	0.1676
4	original_ltv	0.3148	9	original_interest_rate	0.1106
5	upb_rel_change_3m	0.2508	10	loan_purpose	0.1037

Note. Mean absolute TreeSHAP attribution over 20,000 calibration rows; values are on the model's raw log-odds scale.

C.2 Probability-Calibration Diagnostics

Two samples carry these diagnostics. The calibration error is measured on the one-percent hash sample of the calibration window (Appendix B.3) using the raw scores. The Brier and ranking figures are measured on 3,580,221 conformal-partition rows selected by crc32 $\equiv 1 \pmod{50}$, which the calibrator never saw, since an isotonic fit improves in-sample Brier by construction and an in-sample figure would say nothing about transfer.

Table C.3 derives the miscalibration of those raw scores. Weighting the per-bin gaps by occupancy (Naeini et al., 2015) across ten equal-width bins (Guo et al., 2017) returns an expected calibration error of 0.1471, reported as 0.147 in §3.4. The same weighting on the first two columns returns the mean raw score of 0.1583 and the realised rate of 0.0112 that §3.4 sets against each other.

Table C.3: Raw-score reliability on the calibration window, ten equal-width bins

Bin	Mean predicted	Realised rate	Gap	Rows
1	0.0363	0.0013	0.0350	1,074,922
2	0.1432	0.0036	0.1396	262,346
3	0.2461	0.0070	0.2391	140,655
4	0.3471	0.0107	0.3364	94,556
5	0.4473	0.0161	0.4312	71,871
6	0.5468	0.0192	0.5276	51,463
7	0.6462	0.0282	0.6180	34,598
8	0.7461	0.0467	0.6994	21,243
9	0.8502	0.0931	0.7571	16,484
10	0.9587	0.3474	0.6113	29,095
Total	0.1583	0.0112	0.1471	1,797,233

Note. Raw scores; bins are equal-width on $[0, 1]$. The total row reports occupancy-weighted means for Mean predicted, Realised rate and Gap; Rows is the sum.

The calibrator is a pool-adjacent-violators fit (Ayer et al., 1955; Barlow et al., 1972) on the 17.9 million `iso_fit` rows of Appendix B.3, retaining 516 interpolation threshold points. The fitted map is flat on pooled regions and linearly interpolated between retained thresholds; those flat regions create the score atoms analysed in Appendix A.1. Out of sample it reduces the Brier score from 0.064263 to 0.008616, a reduction of 86.6%, and moves AUROC by -0.000060. Table C.4 reads the fitted map at seven points.

Table C.4: Isotonic map at seven raw-score points

Raw score	0.01	0.05	0.10	0.30	0.50	0.70	0.90
Calibrated	0.000363	0.001440	0.002985	0.009569	0.017836	0.035750	0.137322

Note. An illustrative lookup through the fitted map, not the map itself.

C.3 Cross-Model Discrimination and Logistic Benchmark Specification

The benchmark is fitted with balanced class weights by a limited-memory quasi-Newton solver (Liu & Nocedal, 1989) on the 8,303,131 rows of 1999–2003 within the five-percent loan-level sample of Appendix B.3, holding out that sample's 4,081,936 rows of 2004. The L_2 penalty is chosen on those rows by validation AUROC from six values spanning five decades, each candidate fitted on a three-percent subsample of the fitting rows; the selected value is $C = 10^{-4}$, the most strongly penalised end of the grid, across which AUROC varies by only 0.004.

Two input treatments differ from the primary pipeline. All three integer categoricals of §3.3 - vintage year and quarter, and the unit count - enter as standardised numerics, not the origination vintage alone (§3.4). And because the benchmark implementation does not accept numeric nulls natively, every numeric null is imputed at its fit-window median. Existing disclosure indicators among the twenty-nine preserve missingness information for the main origination measures; the benchmark adds one further indicator for the three-month balance change. The design matrix is therefore twenty-five columns passed through the median-impute/standardise pipeline (Hastie et al., 2009, §3.4.1) - the twenty-four non-string inputs plus that indicator - and twenty dummies from the five string categoricals, first level dropped. This results in forty-five columns.

It carries its own isotonic calibrator, fitted under the same `crc32` residue rule (Appendix B.3) on 892,325 calibration-window rows, and its own conformal partition of the remaining 8,080,478. Its SCP threshold covers 92.94% of its own conformal-calibration rows, versus 90.30% for the primary SCP (Table A.1), so the threshold-tie effect of Appendix A.1 is considerably larger here. Table C.5 displays the two scorers side by side.

Table C.5: Discrimination by window, both scorers

		LightGBM		Logistic			Unseen
Window	Base rate (%)	AUROC	AP	AUROC	AP	Δ AUROC	vintage (%)
Calibration	1.11	0.9140	0.3361	0.9089	0.2865	-0.0052	16.3
Crisis	2.53	0.8833	0.3494	0.8810	0.2961	-0.0023	56.4
Normal	1.52	0.8723	0.3135	0.8724	0.2373	$+0.0000$	89.7
COVID	1.68	0.8262	0.2114	0.8216	0.1500	-0.0046	97.6
Rate hike	1.04	0.8719	0.2508	0.8638	0.1403	-0.0081	98.6

Note. Metrics use the respective 2% (LGBM) and 5% (logit) samples; base rates use the full windows. Unseen vintage denotes vintages absent from the 1999–2004 train split.

D Conformal and Evaluation Implementation

D.1 Calibration Thresholds and Mondrian Partitions

Every threshold is the $\lceil (n+1)(1-\alpha) \rceil$-th smallest score of its own calibration set at $\alpha = 0.10$, taken over the 161,421,966 conformal rows of §3.4: the whole partition for the two global methods, the rows of each cell for Mondrian. Each of the four partitions exhausts those rows exactly. Under the exchangeability condition of §2.1, the $1-\alpha$ coverage guarantee attaches per cell (Vovk, 2012, Prop. 3); the tie-free upper bound of Appendix A.1 does not apply to the tied isotonic scores used here. Cells below a thousand calibration rows are separately flagged rather than pooled or dropped, too few rows leaving a threshold unstable rather than invalid. The floor sits an order of magnitude above the informal hundred-example benchmark in Vovk (2012, p. 489) for avoiding very small categories. None of the twenty-seven cells is flagged, the smallest being the 17,312 rows of the LTV sentinel.

The tie structure of Appendix A.1 recurs in every cell: each threshold falls on an atom straddling the target level, so the coverage column stands at the top of the jump that atom creates and exceeds nominal by at most that cell's tie mass. The uniform draw of the APS score (3.3) is seeded per split and observation year, so a re-run reproduces its threshold exactly. Table D.1 reports all four partitions.

Table D.1: Conformal thresholds on the calibration window

Method / partition cell	Calibration rows	$\hat{q}$	$\hat{q} \div \hat{q}_{\mathrm{SCP}}$	Coverage
Global thresholds				
Split conformal	161,421,966	0.016224	1.00	0.903027
Adaptive prediction sets	161,421,966	0.899951	—	0.900000
Mondrian, FICO × LTV (primary)				
Prime × Low	90,850,717	0.003264	0.20	0.901371
Prime × Moderate	7,827,469	0.012462	0.77	0.905145
Prime × High	421,552	0.017995	1.11	0.900501
Near-Prime × Low	42,447,151	0.017836	1.10	0.900103
Near-Prime × Moderate	8,561,321	0.054137	3.34	0.900657
Near-Prime × High	500,659	0.070707	4.36	0.900949
Subprime × Low	7,963,521	0.169072	10.42	0.901254
Subprime × Moderate	2,199,876	0.383388	23.63	0.901370
Subprime × High	93,430	0.420103	25.89	0.900300
Sentinel	556,270	0.018528	1.14	0.900334
Mondrian, FICO tier alone				
Prime	99,112,743	0.004539	0.28	0.900244
Near-Prime	51,512,885	0.024451	1.51	0.908834
Subprime	10,257,284	0.234527	14.46	0.900913
Sentinel	539,054	0.019110	1.18	0.913613

Table D.1, continued

Method / partition cell	Calibration rows	$\hat{q}$	$\hat{q} \div \hat{q}_{\text{SCP}}$	Coverage
Mondrian, LTV bucket alone				
Low	141,710,631	0.012462	0.77	0.903269
Moderate	18,674,053	0.049727	3.07	0.900735
High	1,019,970	0.059754	3.68	0.906399
Sentinel	17,312	0.007393	0.46	0.908387
Mondrian, census division alone				
E North Central	32,288,044	0.018528	1.14	0.900303
E South Central	7,452,812	0.024451	1.51	0.907619
Middle Atlantic	18,573,903	0.015177	0.94	0.902464
Mountain	12,738,736	0.015665	0.97	0.901932
New England	7,712,941	0.010253	0.63	0.900028
Pacific	22,486,396	0.008572	0.53	0.902600
South Atlantic	33,445,238	0.017104	1.05	0.902631
W North Central	14,973,030	0.015177	0.94	0.902144
W South Central	11,750,866	0.024655	1.52	0.900169

Note. Coverage is evaluated on the same calibration rows used to estimate each threshold and is therefore a calibration diagnostic, not an out-of-sample result. The ratio is omitted for APS because it uses a different score scale (3.3).

D.2 Adaptive Conformal Configuration

Both adaptive procedures keep the calibration reference fixed and adapt only their working miscoverage levels: a single α_t for ACI and expert levels α_t^i whose weighted combination is used by DtACI. For an interior working level, the threshold is read at rank $k_t = \lceil (n+1)(1-\alpha_t) \rceil$ from the same frozen, sorted calibration scores that give the split-conformal threshold. The implementation returns $+\infty$ when the adjusted rank exceeds n and $-\infty$ when $1 - \alpha_t \leq 0$. The sequence runs chronologically across all four test windows without resetting at the regime boundaries.

The monthly cadence of §3.5 is a deployment adaptation. The published ACI and DtACI procedures update per observation, whereas this application aggregates row-level misses to realised calendar-month miscoverage and updates once per observed month, yielding 196 steps. The batched ACI update is clipped to $[0, 1]$ as a deployment safeguard; the published recursion is unclipped and instead remains in $[-\gamma, 1+\gamma]$ (Gibbs & Candès, 2021, Lem. 4.1). DtACI uses the deterministic weighted-average aggregation of Gibbs and Candès (2024, §3.4) as its aggregation template, but at monthly resolution each expert is evaluated at its own threshold and its pinball-loss argument is that expert's realised monthly miscoverage rather than the paper's shared per-observation β_t. The expert weights are then updated multiplicatively with the uniform mix-in. This is therefore an empirical deployment adaptation rather than a direct implementation of the published DtACI guarantees. Table D.2 lists the settings.

Table D.2: Adaptive conformal settings

Setting	Value	Basis
Initial level α_0	0.10	nominal target
Update frequency	monthly, 196 steps	deployment adaptation
err_t	month t miscoverage rate	deployment adaptation
Clipping	$\alpha_t, \alpha_t^i \in [0, 1]$	post-update safeguard; recursion unclipped
Reference distribution	sorted cp_cal scores, frozen	design
ACI step sizes γ	$\{0.005, 0.01, 0.02, 0.05\}$	sensitivity grid (§3.5)
DtACI experts K	500	expert pool
DtACI γ grid	geometric on $[0.001, 0.5]$	expert pool
DtACI initial weights	uniform, $1/K = 0.002$	no prior over experts
DtACI learning rate η	3.2343	$\sqrt{(\ln 2K\|I\| + 1)/(\|I\|(1-\alpha)^2\alpha^2/3)}$
DtACI mix-in σ	0.001	$1/(2\|I\|)$
DtACI tuning reference $\|I\|$	500	paper's experimental reference length

Note. The value $|I| = 500$ is borrowed from the experimental tuning choice of Gibbs and Candès (2024) only to instantiate $\sigma = 1/(2|I|)$ and their fixed heuristic approximation for η; it exceeds the 196-step monthly horizon and is not an observed local interval here. No published DtACI guarantee is claimed: the monthly loss replaces the paper's shared per-observation β_t with expert-specific realised monthly miscoverage, the sufficient grid condition of Theorem 4, $\gamma_K \geq \sqrt{1 + 1/|I|} \approx 1.001$, is not met, and the exact long-run conclusion of Theorem 6 requires $\eta_t \to 0$ and $\sigma_t \to 0$ rather than fixed values.

D.3 Metric Definitions and Inference Conventions

Coverage strata. Temporal coverage is a twelve-month trailing window computed from monthly covered and total counts, not by re-averaging monthly rates, so months contribute in proportion to their size. The window restarts at each test-window boundary, so the first eleven months of each are partial by construction, and the adaptive methods' rolling coverage excludes the preceding regime even though their working level (§3.5) carries across it. The score-decile profile cuts each window into ten equal-frequency bins of the calibrated probability, duplicate edges collapsed where the isotonic map ties them.

Miscoverage models. The diagnostics below the predefined cells fit a gradient-boosted classifier to the miscoverage indicator. Braun et al. (2025) show that, under a proper loss, the exact target condition $P(M = 1 \mid X) = \alpha$ makes the constant predictor α Bayes-optimal. Zhou et al. (2026) likewise cast local coverage assessment as supervised reliability estimation. Here AUROC is used only as a descriptive measure of the separability of miscovered and covered observations, not as the excess-risk or validity-index summaries proposed in those papers. The score-only specification carries the calibrated probability alone. The full one adds the borrower-risk cell, its two margins, census division, observation year and regime. Discrimination is evaluated leaving one regime out at a time, the early-stopping slice carved from the training regimes, so the reported figure is transfer to an unseen regime rather than in-sample fit.

Concentration and pockets. The concentration curve (3.7) comes from a separate pooled model, fitted on the observation years 2007–2018, 2020 and 2021 and evaluated on 2019, 2022 and 2023, so predicted risk is out of sample throughout. The area is taken by the trapezoid rule over 201 grid points on the ranking by decreasing predicted risk. That year-based holdout is not a clean chronological cut, and it bounds the diagnostic's scope: only the normal and rate-hike windows contribute evaluation-year rows, so concentration areas are reported for those two alone, while the leave-one-regime-out discrimination above covers all four. The curve is a cumulative accuracy profile in the sense standard to credit-risk validation (Engelmann et al., 2003), with realised miscoverage in place of default and the reliability model's predicted risk in place of the rating score. It shares the cumulative-share geometry of the empirical Lorenz curve (Gastwirth, 1972) but orders by predicted risk rather than by the accumulated quantity itself. Its area $A = \int_0^1 \mathrm{Conc}(f)\,df$ is therefore used here as a rank-based concentration summary, not as an ordinary Gini index. $1/2$ is the diagonal no-concentration reference, and larger values indicate that realised miscoverage is increasingly concentrated near the top of the predicted-risk ranking. The pockets are the leaves of a depth-three Gini-split tree (Breiman et al., 1984) on the same features less the regime label, which is excluded so the tree cannot split directly on the pre-assigned regime categories, with a minimum leaf size of half a percent of the audited rows, fitted in sample on all of them.

Matched budget. The comparison (3.8) reads capture off the calibrated probability ranked in decreasing order at the review budget the conformal rule itself sets.

Population stability index. The ten equal-frequency bins of (2.6) take their boundaries from the deciles of the calibration-window reference distribution, the outer edges opened to $\pm\infty$ so that every deployment observation is assigned and empty bins floored at 10^{-8} before the logarithm. Duplicate decile boundaries are collapsed, which on a discrete score can drop the effective bin count below ten and would make the value incomparable with the ten-bin convention of §2.5. The effective count is ten in all four test windows. The Jeffreys equivalence of §2.5 is a property of this fixed binning rather than of the index, since exchanging the two samples' roles would move the cut-points (Yurdakul & Naranjo, 2020).

The conventional bands of §2.5 are rules of thumb with no stated error rate. Yurdakul and Naranjo (2020) derive an asymptotic reference for two independent multinomial samples over B fixed categories: under the null of no shift, $(1/n + 1/m)^{-1}\mathrm{PSI}$ is approximately χ^2_{B-1}. Under that approximation, the null mean is $(1/n + 1/m)(B - 1)$, and the level-α reference threshold is

$$\text{PSI} > \left(\frac{1}{n} + \frac{1}{m}\right) F^{-1}_{\chi^2_{B-1}}(1-\alpha). \tag{D.1}$$

Table D.3 evaluates this reference at $\alpha = 0.05$ and $B = 10$, for which $F^{-1}_{\chi^2_9}(0.95) = 16.919$. Every realised PSI exceeds the corresponding 5% reference threshold by orders of magnitude, including the crisis value of 0.0062 that the conventional bands classify as negligible. This is not treated as a calibrated hypothesis test for the mortgage panel. Repeated loan-month observations violate the independent-multinomial sampling model, and the decile cut-points are estimated from the calibration reference rather than fixed ex ante. PSI is therefore interpreted descriptively.

Table D.3: Asymptotic no-shift reference for the PSI

Test window	m (millions)	Approx. $\mathbb{E}[\text{PSI}]$ under H_0	5% reference	Realised PSI	0.10÷ reference
Crisis 2007–12	12.7	3.2×10^{-6}	6.07×10^{-6}	0.0062	16,500
Normal 2013–19	15.1	3.1×10^{-6}	5.86×10^{-6}	0.0891	17,100
COVID 2020–21	4.02	4.8×10^{-6}	8.94×10^{-6}	0.1240	11,200
Rate hike 2022–23	6.12	4.0×10^{-6}	7.50×10^{-6}	0.1988	13,300

Note. $n = 3{,}572{,}364$ is the calibration-window crc32 $\equiv 3 \pmod{50}$ reference; m applies the same hash class to each test window (Appendix B.3).

Interval reporting. Clopper–Pearson intervals (Clopper & Pearson, 1934) are reported in full for marginal coverage in Table D.4 and are not printed beside every finer stratum. No half-width there reaches 0.01 percentage points, so interval exclusion separates nothing at this granularity. The reporting rule is therefore stated once: a stratum in Appendices E to G carries its own interval only where its half-width exceeds half a percentage point, a quarter of the practical-significance band. Among the strata whose counts the run logs record, that binds only on the Subprime $\times$ High cell of the five-percent audit sample, reaching 0.60 percentage points in the rate-hike window and 0.59 in the COVID window.

Table D.4: Clopper–Pearson 95% intervals for marginal coverage

Window	Coverage (%)	CI low	CI high	Deviation (pp)	Material
Split conformal					
Crisis	90.515	90.512	90.517	$+0.515$	no
Normal	94.548	94.547	94.550	$+4.548$	yes
COVID	94.827	94.824	94.830	$+4.827$	yes
Rate hike	96.409	96.407	96.411	$+6.409$	yes
Mondrian, FICO $\times$ LTV					
Crisis	89.503	89.500	89.505	-0.497	no
Normal	94.269	94.267	94.270	$+4.269$	yes
COVID	93.904	93.901	93.907	$+3.904$	yes
Rate hike	95.291	95.289	95.294	$+5.291$	yes

Table D.4, continued

Window	Coverage (%)	CI low	CI high	Deviation (pp)	Material
Adaptive prediction sets					
Crisis	88.994	88.992	88.997	−1.006	no
Normal	89.461	89.459	89.463	−0.539	no
COVID	89.142	89.138	89.147	−0.858	no
Rate hike	89.584	89.581	89.588	−0.416	no
DtACI					
Crisis	90.415	90.413	90.417	+0.415	no
Normal	93.807	93.805	93.809	+3.807	yes
COVID	94.303	94.300	94.307	+4.303	yes
Rate hike	95.982	95.980	95.984	+5.982	yes

D.4 Robustness Design Matrix

Table D.5 is the single authority for which analyses were run under which alternative. Three are carried: the logistic benchmark of §3.4, that same benchmark scored without its isotonic map, and a prepayment-restricted sample that drops loan-months with code-01 payoff/maturity in the forward window (§3.1). None is a full replication of the primary pipeline, and the blanks are as informative as the marks.

Table D.5: Which analyses were run under which pipeline

Analysis	Primary pipeline	Logistic benchmark	Uncalibrated scores	Prepayment-restricted
Split conformal	•	•	•	•
Mondrian, FICO × LTV	•	•		•
Mondrian, three coarse partitions	•			
Adaptive prediction sets	•	•	•	•
ACI, four step sizes	•			
DtACI	•			
Class-conditional and subgroup coverage	•	•	•	•
Score-decile and pocket audit	•			
Drift monitoring	•	•		
Operational routing and matched budget	•	•		

Note. The prepayment restriction is applied to the primary pipeline alone and removes between 6.1% and 24.1% of loan-months depending on the window. Corresponding reported results are in Appendices E.2, E.3, E.11, F.4 and G.5.

E Supplementary Coverage and Reliability Results

E.1 Complete Static-Method Coverage Results

Tables 4.1 and 4.2 report marginal and delinquent-class coverage, the empty-set rate and the subgroup summaries. Table E.1 extends the performing-class rate of Table 4.1 to all three static methods and resolves the remaining set types, whose four shares exhaust each window.

Table E.1: Performing-class coverage and prediction-set composition by method and regime

Method	Regime	$\widehat{\mathrm{Cov}}(y{=}0)$ (%)	$\emptyset$ (%)	$\{0\}$ (%)	$\{1\}$ (%)	$\{0,1\}$ (%)
SCP	Crisis	92.86	8.52	91.48	0.000	0.000
	Normal	96.00	4.82	95.18	0.000	0.000
	COVID	96.45	4.29	95.71	0.000	0.000
	Rate hike	97.42	3.08	96.92	0.000	0.000
Mondrian (FICO × LTV)	Crisis	91.80	9.34	90.62	0.034	0.000
	Normal	95.71	5.01	94.98	0.018	0.000
	COVID	95.50	5.18	94.82	0.005	0.000
	Rate hike	96.29	4.17	95.82	0.003	0.000
APS	Crisis	90.53	9.32	89.15	0.105	1.431
	Normal	90.39	9.52	89.46	0.059	0.960
	COVID	90.37	9.56	89.70	0.025	0.724
	Rate hike	90.30	9.65	89.83	0.020	0.506

Note. Full test windows.

On the calibration window SCP splits its 9.697 percentage points of miscoverage into 9.406 arriving as the empty set and 0.291 as a singleton excluding the realised label. The two-label rate is exactly zero, so the singleton rate is 90.594%. APS on the same rows attains 90% marginal coverage, 90.5867% on the performing class and 37.6046% on the delinquent class, its 10% miscoverage not decomposed by set type. Mondrian on the crossed partition attains 90.12% marginal, its class split not separately recorded. The sentinel cell, which absorbs loan-months with a missing FICO or LTV field (§3.3), is excluded from the cell-level profiles of Appendix E.9 but not from the CovGap and worst-group columns, which apply a thirty-row minimum and no further filter. Under SCP's global threshold it is covered at 88.05% on its 556,270 conformal rows, a 1.95-point shortfall - between the Prime × High and Near-Prime × Low cells, and far from either extreme, so it sets neither column on the calibration window. Its test-window coverage is not separately tabulated, as in the five-percent audit sample, the sentinel cell accounts for only 0.02% - 0.20% of rows across windows.

E.2 Effect of Probability Calibration on Class-Conditional Coverage

The benchmark pipeline of Appendix C.3 is the only one run with and without its isotonic map, so it provides a within-pipeline contrast for the map's incremental effect while holding the benchmark scorer and class ratio fixed. Fitting the split-conformal threshold on raw logistic scores puts it at 0.4347. The same construction on the calibrated scores puts it at 0.0184, a factor of 23.7 lower. The APS threshold moves in the opposite direction but much less, from 0.8504 to 0.8998, its score being a cumulative probability rather than a probability. Table E.2 evaluates both on the four test windows.

Table E.2: Coverage under raw and isotonic-calibrated benchmark scores

		Marginal (%)		$\widehat{\mathrm{Cov}}(y{=}1)$ (%)	
Method	Regime	Raw	Calibrated	Raw	Calibrated
SCP	Crisis	89.12	91.08	58.60	0.33
	Normal	92.51	93.84	57.03	0.80
	COVID	93.18	94.29	42.62	0.14
	Rate hike	94.68	95.80	48.56	1.43
APS	Crisis	88.85	89.04	69.27	27.73
	Normal	88.77	89.51	64.03	28.97
	COVID	88.56	89.19	53.77	15.51
	Rate hike	89.09	89.54	58.24	21.89

Note. The calibrated columns are the benchmark-pipeline results of Table E.11, repeated here for the contrast.

The mechanism is the map's compression of the score support. On the calibration sample the raw scores have mean 0.1773 and standard deviation 0.1997, with 1.6% below 0.01. After the map the mean is 0.0111 and the standard deviation 0.0441, with 82.2% below 0.01. The threshold, a quantile of a distribution that is 98.9% performing-class mass, follows the bulk downwards while the delinquent scores do not, and the class-conditional rate collapses.

E.3 Prepayment-Restricted Sensitivity Analysis

Table E.3 re-evaluates the three static methods after dropping loan-months with code-01 payoff/maturity in the forward window (§3.1). Thresholds are not refitted, so calibration is held fixed.

Table E.3: Reliability after excluding prepaid loan-months

Method	Regime	Δ MC (pp)	$\widehat{\mathrm{Cov}}(y{=}1)$ (%)	CovGap (pp)
SCP	Crisis	-0.992	0.00072	80.13
	Normal	-0.228	0.00022	47.88
	COVID	-0.273	0.00000	33.42
	Rate hike	$+0.060$	0.00007	24.89

Table E.3, continued

Method	Regime	Δ MC (pp)	$\widehat{\text{Cov}}(y=1)$ (%)	CovGap (pp)
Mondrian (FICO $\times$ LTV)	Crisis	-0.749	0.99	6.59
	Normal	-0.019	0.83	7.23
	COVID	-0.122	0.20	4.89
	Rate hike	$+0.154$	0.20	6.20
APS	Crisis	-0.220	29.70	4.38
	Normal	-0.063	29.47	1.48
	COVID	-0.166	17.59	2.20
	Rate hike	$+0.006$	21.41	1.65

Note. Δ MC is restricted minus full-sample marginal coverage. The restriction removes 16.83%, 12.87%, 24.08% and 6.11% of loan-months in window order.

E.4 Alternative Mondrian Partition Results

Each coarse partition is evaluated twice: on the dimension it partitions, where its guarantee attaches, and on the crossed FICO $\times$ LTV grid of Table 4.2, in each case against split conformal.

Table E.4: Coarse Mondrian partitions on their own dimension and on the crossed grid

			Own-dimension CovGap (pp)		Crossed CovGap (pp)
Partition / regime	MC (%)	$\widehat{Cov}(y=1)$ (%)	Mondrian	SCP	Mondrian
FICO tier (3 cells + sentinel)					
Crisis	89.18	0.14	4.89	43.49	42.38
Normal	92.56	0.05	3.10	30.78	14.44
COVID	91.68	0.01	1.80	20.03	20.86
Rate hike	93.30	0.01	3.67	13.70	20.37
LTV bucket (3 cells + sentinel)					
Crisis	90.54	0.0007	0.62	16.78	49.83
Normal	94.87	0.0002	5.63	5.75	35.18
COVID	95.09	0.0000	5.20	5.99	25.08
Rate hike	96.55	0.0001	6.85	7.24	19.24
Census division (9 cells)					
Crisis	90.55	0.0007	1.62	4.88	76.28
Normal	94.33	0.0002	5.31	7.01	51.46
COVID	94.63	0.0000	5.82	6.11	33.95
Rate hike	96.32	0.0001	7.02	7.57	27.04

Note. Own-dimension CovGaps exclude the sentinel; crossed-grid CovGaps include it, as in Table 4.2.

E.5 Coverage by Score Decile

Table E.5 reports the profile Figure 4.2 plots, on the five-percent audit sample: 31,692,753 crisis rows, 37,785,626 normal, 10,079,140 COVID and 15,312,169 rate-hike, each cut into ten equal-frequency bins of the calibrated probability, so a decile holds about a tenth of its window.

Table E.5: Coverage by decile of the calibrated probability, method and regime

Decile	Crisis (%)	Normal (%)	COVID (%)	Rate hike (%)
SCP				
1 (safest)	99.95	99.89	99.80	99.93
2	99.94	99.84	99.73	99.97
3	99.73	99.86	99.77	99.93
4	99.89	99.79	99.20	99.77
5	99.07	99.10	99.81	99.92
6	99.06	99.10	98.46	99.32
7	98.92	99.23	99.13	99.39
8	97.71	98.99	98.44	99.23
9	96.40	98.48	98.30	98.96
10 (riskiest)	14.25	50.53	50.45	65.58
Mondrian, FICO $\times$ LTV				
7	86.78	99.23	99.13	99.39
8	73.97	90.74	95.56	99.23
9	85.53	88.26	86.94	85.69
10	49.16	65.46	57.31	67.29
APS				
1 (safest)	89.99	89.94	89.78	89.95
2	89.96	89.93	89.84	90.00
3	89.87	89.94	89.87	90.01
4	90.00	89.91	89.36	89.87
5	89.29	89.27	89.89	90.04
6	89.36	89.34	88.72	89.49
7	89.31	89.51	89.39	89.60
8	88.43	89.36	88.83	89.54
9	87.67	89.18	88.92	89.52
10 (riskiest)	85.88	87.66	86.07	87.63

Note. Omitted Mondrian deciles coincide with split conformal at the reported precision.

E.6 Miscoverage-Prediction Models

Table E.6 reports the discrimination of the classifiers specified in Appendix D.3, each trained on three regimes and evaluated on the fourth.

Table E.6: Miscoverage-model AUROC, leaving one regime out

	SCP		Mondrian		APS	
Held-out regime	Score	Full	Score	Full	Score	Full
Crisis	0.9973	0.9972	0.9369	0.9962	0.5733	0.5727
Normal	0.9953	0.9957	0.9560	0.9950	0.5487	0.5474
COVID	0.9921	0.9930	0.9575	0.9933	0.5585	0.5561
Rate hike	0.9949	0.9948	0.9615	0.9958	0.5381	0.5350
Mean	0.9949	0.9952	0.9530	0.9951	0.5546	0.5528

Note. Five-percent audit sample.

The pooled model's split gains attribute the same pattern. The calibrated probability carries 93.5% of the gain under SCP and 96.2% under APS, against 70.0% under Mondrian; the FICO $\times$ LTV cell and its two margins together carry 5.6%, 1.2% and 29.3%. The regime label carries under 0.5% for every method.

E.7 Miscoverage-Concentration Analysis

Table E.7 reports the concentration area (3.7) and three points on the curve behind it. The pooled model, its allocation and the restriction to the 2019 and 2022–23 evaluation rows are in Appendix D.3.

Table E.7: Concentration of miscoverage under a predicted-risk ranking

			Miscoverage captured (%) in the top		
Method	Regime	Area	5%	10%	20%
SCP	Normal	0.955	82.08	90.47	96.04
	Rate hike	0.977	95.36	97.28	99.05
Mondrian	Normal	0.955	78.35	91.41	96.48
	Rate hike	0.973	95.40	97.92	99.29
APS	Normal	0.579	15.44	21.61	32.09
	Rate hike	0.533	9.48	14.67	24.64

Note. An area of 0.5 denotes no concentration.

E.8 Local Reliability Pockets

The depth-three trees of Appendix D.3 retain six leaves under SCP and eight under Mondrian and APS, at in-sample AUROCs of 0.973, 0.962 and 0.524. Table E.8 lists every leaf the one-sided screen returns.

Table E.8: Depth-three reliability pockets on the pooled audit rows

Method	Leaf	Support (%)	Miscov. (%)	Mean $\hat{p}$	Dominant cell (share)	Dominant regime (share)
SCP	9	5.21	100.00	0.0806	Near-Prime $\times$ Low (30.8)	Crisis (49.6)
	10	0.52	99.997	0.5138	Near-Prime $\times$ Low (31.4)	Crisis (51.7)
	6	1.06	6.31	0.0046	Near-Prime $\times$ Low (38.7)	Normal (39.2)
	7	15.27	2.12	0.0084	Near-Prime $\times$ Low (38.7)	Crisis (41.4)
	4	31.16	0.89	0.0023	Prime $\times$ Low (45.6)	Normal (40.1)
	3	46.79	0.15	0.0007	Prime $\times$ Low (86.0)	Normal (41.2)
Mondrian	14	2.23	99.98	0.1341	Near-Prime $\times$ Low (73.3)	Crisis (47.4)
	11	0.50	71.72	0.3400	Subprime $\times$ Low (68.2)	Crisis (53.0)
	13	1.51	54.51	0.1142	Near-Prime $\times$ Mod. (62.3)	Crisis (45.6)
	7	6.50	47.27	0.0062	Prime $\times$ Low (44.3)	Crisis (39.2)
	10	1.04	6.48	0.0359	Subprime $\times$ Low (59.3)	Crisis (58.5)
	6	13.88	2.29	0.0082	Near-Prime $\times$ Low (56.1)	Crisis (40.8)
	4	27.56	0.94	0.0021	Prime $\times$ Low (48.3)	Normal (40.1)
	3	46.79	0.15	0.0007	Prime $\times$ Low (86.0)	Normal (41.2)

Table E.8, continued

Method	Leaf	Support (%)	Miscov. (%)	Mean $\hat{p}$	Dominant cell (share)	Dominant regime (share)
APS	11	1.66	18.81	0.1153	Near-Prime × Low (34.6)	Crisis (42.7)
	6	1.06	15.26	0.0046	Near-Prime × Low (38.7)	Normal (39.2)
	10	1.10	14.19	0.0338	Near-Prime × Low (27.5)	Crisis (54.1)
	13	0.52	12.10	0.2933	Near-Prime × Low (35.5)	Crisis (45.9)
	7	17.19	11.32	0.0096	Near-Prime × Low (37.5)	Crisis (42.8)

Note. The screen reports at most five leaves per side; APS has no screened overcoverage leaves.

E.9 Cell-Level Miscoverage Profiles

Table E.9 reports miscoverage in each observed non-sentinel FICO × LTV cell, the stratification the pockets of Appendix E.8 cut across.

Table E.9: Miscoverage by borrower-risk cell, method and regime

Cell	Rows	SCP (%)	Mondrian (%)	APS (%)	SCP $\emptyset$ (%)
Crisis 2007–12					
Prime × Low	18,434,424	1.75	9.49	10.49	1.20
Prime × Moderate	2,060,556	5.53	7.85	11.09	4.30
Prime × High	307,998	6.16	5.59	11.42	4.37
Near-Prime × Low	7,369,521	13.36	12.37	11.66	11.79
Near-Prime × Moderate	1,532,374	36.18	13.05	12.16	34.66
Near-Prime × High	211,554	42.45	13.40	13.01	40.40
Subprime × Low	1,328,488	46.00	15.11	12.33	44.56
Subprime × Moderate	332,791	76.84	13.56	12.14	76.20
Subprime × High	52,057	88.94	16.07	14.89	88.30
Normal 2013–19					
Prime × Low	20,984,784	1.34	4.97	10.32	0.96
Prime × Moderate	4,796,869	2.37	2.95	10.43	1.77
Prime × High	1,480,544	2.79	2.61	10.59	1.96
Near-Prime × Low	6,453,040	9.42	8.93	10.99	8.36
Near-Prime × Moderate	1,898,860	16.60	7.96	11.05	15.37
Near-Prime × High	739,440	16.32	6.71	11.05	14.91
Subprime × Low	956,356	33.88	12.92	11.28	32.61
Subprime × Moderate	276,503	54.37	9.27	10.75	53.39
Subprime × High	177,984	57.37	6.59	10.83	56.56
COVID 2020–21					
Prime × Low	5,772,056	2.02	5.28	10.58	1.46
Prime × Moderate	1,514,130	4.13	4.89	10.87	3.22
Prime × High	292,499	5.85	5.52	11.21	4.60
Near-Prime × Low	1,600,217	8.96	8.57	11.43	7.58
Near-Prime × Moderate	516,138	16.20	8.56	11.58	14.53
Near-Prime × High	146,616	18.03	9.03	11.64	16.13
Subprime × Low	165,897	26.04	9.97	11.90	24.22
Subprime × Moderate	41,924	42.26	7.80	11.45	40.90
Subprime × High	26,587	39.65	5.60	10.59	38.40

Table E.9, continued

Cell	Rows	SCP (%)	Mondrian (%)	APS (%)	SCP ∅ (%)
Rate hike 2022–23					
Prime × Low	8,920,184	1.17	4.18	10.20	0.90
Prime × Moderate	2,273,204	2.91	3.83	10.35	2.40
Prime × High	353,779	4.38	4.03	10.47	3.72
Near-Prime × Low	2,574,935	6.71	6.38	10.92	5.72
Near-Prime × Moderate	729,756	13.35	6.59	11.10	12.15
Near-Prime × High	157,618	14.59	6.91	10.82	13.43
Subprime × Low	232,937	20.85	7.77	11.34	19.41
Subprime × Moderate	42,739	35.12	6.33	10.66	34.00
Subprime × High	23,435	31.51	3.37	9.85	30.80

Note. Five-percent audit sample; Mondrian uses the FICO × LTV partition. Under the interval rule of Appendix D.3, SCP in Subprime × High is 39.65% [39.06, 40.24] in COVID and 31.51% [30.91, 32.11] in the rate-hike window.

The same cells on the calibration window, under the single global threshold from which they are drawn, give the profile §4.1 quotes: Prime × Low 99.03%, Prime × Moderate 94.33%, Prime × High 87.62%, Near-Prime × Low 88.83%, Near-Prime × Moderate 60.26%, Near-Prime × High 49.39%, Subprime × Low 51.38%, Subprime × Moderate 15.43% and Subprime × High 7.42%. APS on the largest of these covers at 89.99% against SCP's 99.03%, its delinquent-class rate inside the cell rising from 0.00% to 16.08%. Mondrian's per-cell calibration coverage is in Appendix D.1.

APS's residual failure concentrates on the score axis rather than on the cells. Splitting each window at a calibrated probability of five percent, its miscoverage runs 10.84%, 10.45%, 10.71% and 10.34% below the cut against 15.71%, 14.34%, 17.52% and 15.45% above it, on 1.11, 0.90, 0.23 and 0.23 million rows respectively - between 1.5% and 3.5% of each window.

E.10 Adaptive Coverage and Rolling Reliability

Table E.10 reports the full adaptive grid. Only DtACI appears in Table 4.2. The four ACI step sizes are the sensitivity check of §3.5.

Table E.10: Adaptive methods: per-regime and long-run coverage, and the adapted level

Method	Crisis (%)	Normal (%)	COVID (%)	Rate hike (%)	Long run (%)	Final α_t	Target	$\hat{q}_t$ range
ACI $\gamma = 0.005$	90.51	94.09	94.07	95.73	93.16	0.1288	0.871	0.0132–0.0162
ACI $\gamma = 0.01$	90.50	93.72	93.44	94.90	92.80	0.1514	0.849	0.0109–0.0162
ACI $\gamma = 0.02$	90.48	93.01	92.41	93.92	92.25	0.1829	0.817	0.0090–0.0164
ACI $\gamma = 0.05$	90.45	91.69	90.90	92.16	91.27	0.2193	0.781	0.0069–0.0171
DtACI	90.42	93.81	94.30	95.98	93.08	0.1128	0.887	0.0142–0.0164

Note. Target is $1 - \alpha_t$ at the final step.

Read monthly, only the crisis window breaches nominal coverage, and it does so under SCP, Mondrian and DtACI. Over the 61 crisis months for which a full twelve-month window exists, rolling coverage sits below nominal for 27 consecutive months under SCP, 26 under DtACI and 46 under Mondrian, reaching minima of 89.04%, 89.13% and 88.00%; single-month minima are 88.66%, 88.85% and 87.40%. In the normal, COVID and rate-hike windows no full window of any of the three falls below nominal. APS is below nominal in every month of every window, consistent with its marginal undercoverage rather than an onset-specific break.

E.11 Static-Method Robustness Across Scorers and Samples

Table E.11 re-runs the three static methods on the isotonic-calibrated logistic benchmark.

Table E.11: Static-method reliability under the logistic benchmark

Method	Regime	MC (%)	95% CI	$\widehat{\text{Cov}}(y=1)$ (%)	Empty (%)	CovGap (pp)
SCP	Crisis	91.08	[91.07, 91.09]	0.33	7.89	76.54
	Normal	93.84	[93.84, 93.85]	0.80	5.48	69.85
	COVID	94.29	[94.28, 94.31]	0.14	4.76	70.04
	Rate hike	95.80	[95.79, 95.81]	1.43	3.63	67.48
Mondrian	Crisis	90.33	[90.32, 90.34]	0.60	8.48	7.29
(FICO $\times$ LTV)	Normal	94.09	[94.08, 94.09]	1.12	5.16	6.20
	COVID	93.41	[93.39, 93.42]	0.22	5.67	4.79
	Rate hike	94.55	[94.54, 94.57]	1.54	4.88	5.65
APS	Crisis	89.04	[89.03, 89.05]	27.73	9.19	2.96
	Normal	89.51	[89.50, 89.52]	28.97	9.41	2.43
	COVID	89.19	[89.17, 89.20]	15.51	9.47	2.69
	Rate hike	89.54	[89.52, 89.55]	21.89	9.58	2.76

Note. Clopper-Pearson intervals use the five-percent loan-level sample. The prepayment-restricted sample is in Appendix E.3.

Three differences from the primary pipeline bear on the attribution. The benchmark's threshold is 0.0184 against 0.0162, and its cell-wise thresholds span a factor of 83.1 against 128.7, consistent with less aggressive cross-cell reordering. The larger SCP threshold-tie effect raises its conformal-calibration coverage to 92.94% (Appendix C.3). On those rows it covers the delinquent class at 0.1024% under SCP and 32.99% under APS. And SCP's subgroup CovGap does not narrow across the regimes, staying between 67.48 and 76.54 points where the primary pipeline falls from 79.69 to 24.94.

F Drift Monitoring and Regime Diagnostics

F.1 Score-Distribution Diagnostics by Regime

Table F.1 sets the stability index against the two properties of the score distribution it is meant to summarise: where the mass sits on average, and how much of it stands above the conformal threshold.

Table F.1: Score-distribution movement and the stability index, by test window

Window	Mean score Δ (%)	Above-threshold share, Δ (pp)	PSI
Crisis 2007–12	$+65.8$	-0.26	0.0062
Normal 2013–19	$+5.8$	-4.28	0.0891
COVID 2020–21	$+15.8$	-4.53	0.1240
Rate hike 2022–23	-23.4	-6.09	0.1988

Note. Changes are relative to calibration (mean realised-label SCP score 0.017569; above-threshold share 9.70%). Test-window score diagnostics use a separate two-percent draw; PSI uses the reference sample of Table B.3.

The crisis pairs the largest mean-score rise of the four with the smallest index, and the pairing resolves in the minority class. Its positive rate is more than double the calibration rate (Table 3.1), virtually every delinquent score lies above the threshold (§4.1), and the performing class's own above-threshold share falls over the same move, from 8.69% to 7.14% (Tables 4.1 and E.1). Carrying the calibration class means of §4.1 at the crisis positive rate reproduces roughly nine-tenths of the realised mean shift, so most of the mean shift is compositional. That accounting holds the class means fixed.

The second column helps explain why this barely reaches the index. PSI bins are fixed from calibration-sample deciles and not from the conformal threshold itself (Appendix D.3), so threshold exceedance and top-bin occupancy are not identical quantities. The above-threshold share changes by only -0.26 points in the crisis against -6.09 in the rate-hike window, even as the crisis mean score rises by 65.8%. An index defined on bin occupancies is blind to movement within bins. The crisis pattern is therefore consistent with turnover in the high-score region while fixed-bin occupancies barely redistribute.

F.2 Regime Classification and Supporting Evidence

Table F.2 records what each design label of §3.2 predicts, what the evidence shows and where the assignment is qualified. The score statistics behind the middle column are in Table F.1.

Table F.2: A priori regime labels against the realised evidence

Window (label)	Signature the label predicts	Realised evidence	Assessment
Crisis (tail shift)	Binned bulk near-stationary, tail mass moving far above the threshold, positive rate well above calibration	Lowest index and largest mean-score rise of the four; positive rate +1.42 pp, the largest move of the four	Consistent; the split between the index and the mean shift is itself the signature
Normal (stable drift)	Index, mean shift and positive rate all close to calibration	Second-lowest index and smallest mean-score rise of the four; positive rate +0.41 pp	Consistent in aggregate and the most qualified of the four: the COVID-reaching label share of Appendix B.2 leaves the window internally inhomogeneous
COVID (structural shift)	Score distribution displaced and the score-to-label relation altered by policy rather than by credit alone	Index inside the investigation band, mean-score rise between the normal and crisis values; positive rate +0.57 pp, alongside the documented forbearance intervention (§3.2)	Consistent, but resting on the documented intervention rather than on the index
Rate hike (composition shift)	Score distribution compressed downward, lower positive rate	Highest index of the four and the only negative mean-score shift; positive rate −0.06 pp, the only window below calibration	Consistent

Note. Positive-rate changes use full-window rows relative to the 1.1075% calibration rate (Table B.2).

No assignment is contradicted. What the table establishes is consistency with the a priori labels, not identification of a mechanism: four windows and three descriptive statistics cannot separate competing explanations.

F.3 Feature-Level Population Stability Indices

The label-free variant of §3.6 runs on four numeric origination characteristics - credit score, loan-to-value ratio, note rate and debt-to-income ratio - against the same calibration window (Table F.3).

Table F.3: Feature-level stability indices against the calibration window

Origination characteristic	Crisis	Normal	COVID	Rate hike
Credit score	0.0302	0.1513	0.2103	0.2190
Loan-to-value ratio	0.0075	0.1146	0.1278	0.1024
Note rate	0.1364	2.4898	3.9945	3.9695
Debt-to-income ratio	0.0111	0.0823	0.1594	0.1734

Note. Active loan-months only; sentinels and zero note rates dropped before binning.

The crisis carries the smallest index of the four windows on every characteristic, so the endpoint dissociation survives the removal of the label. A monitor reading no outcome at all still rates the least reliable window the most stable. The full four-window inversion of the reliability ordering holds on the credit score and the debt-to-income ratio alone. The leverage and note-rate columns preserve the crisis endpoint but order their three later windows differently, so what replicates label-free is that endpoint. The note rate is the one characteristic of the four to leave the negligible band in the crisis. The three categorical origination fields - loan purpose, occupancy status and property type - again place the crisis smallest, at no more than 0.0115, and again disagree on the windows after it.

F.4 PSI and Coverage Under the Logistic Benchmark

Table F.4 repeats the score-level drift diagnostics on the isotonic-calibrated logistic benchmark. Its coverage and CovGap results are in Table E.11.

Table F.4: Stability index and score movement under the logistic benchmark

Window	PSI	Band	Mean score Δ (%)
Crisis	0.0084	negligible	$+75.12$
Normal	0.1171	investigation	$+16.74$
COVID	0.1669	investigation	$+23.97$
Rate hike	0.2565	action	-9.95

Note. Reference mean split-conformal score: 0.017976.

Every index exceeds its primary-pipeline counterpart in Table F.1 by near-constant factors of 1.29 to 1.35. That uniform excess is consistent with the vintage note of §3.4, but scorer, preprocessing, calibration map and diagnostic sample change jointly, so its source is not isolated. The mean-score column moves in the same direction as the primary pipeline's in every window, the rate-hike compression alone being shallower.

Read against the split-conformal subgroup CovGaps of Table E.11, the ordering replicates at the endpoints and not between them. The crisis carries the lowest index and the widest CovGap, the rate-hike window the highest index and the narrowest, while the normal and COVID windows exchange places on a margin of 0.19 points of CovGap. The banding shifts as well: two of the four windows change class (Table F.5).

F.5 Monitoring Threshold Classification

Table F.5 applies the conventional bands of §2.5 to the indices of Tables F.1 and F.4, and sets the resulting classification against the reliability ordering it is taken to track.

Table F.5: Band classification and the reliability ordering it is read against

	Band		Rank (1 = most favourable)			Marginal dev.
Window	Primary	Benchmark	PSI	CovGap	WGC	material
Crisis	negligible	negligible	1	4	4	no
Normal	negligible	investigation	2	3	3	yes
COVID	investigation	investigation	3	2	2	yes
Rate hike	investigation	action	4	1	1	yes

Note. Reliability ranks use the split-conformal results of Table 4.1.

Nothing reaches the action band under the primary scorer. A monitor operating these cutoffs would therefore open a score-level investigation in two of the four windows and never in the crisis, and the two it does open are the two best covered across subgroups. The rank columns make that reversal exact, summing to five in every row against the index column. On these split-conformal results CovGap is the worst-group shortfall, so the two rank identically. Under the benchmark the rate-hike window does reach the action band, and it is that pipeline's narrowest-CovGap window (Table E.11). The index orders the four marginal deviations exactly, but the bands do not reproduce the materiality split. The normal window's marginal deviation is material by the criterion of §3.6 while its index stays inside the negligible band, so of the three windows carrying material marginal overcoverage the bands flag two.

G Supplementary Operational Routing Results

G.1 Routing Outcomes by Borrower-Risk Cell

Table G.1 resolves the two cells §4.4 quotes into the full nine-cell grid. The cell-wise thresholds behind the Mondrian columns are in Appendix D.1.

Table G.1: Review budget and its delinquency rate by borrower-risk cell and method

Cell	Pos. rate (%)	SCP Budget (%)	SCP Prec. (%)	Mondrian Budget (%)	Mondrian Prec. (%)	APS Budget (%)	APS Prec. (%)
Prime × Low	0.70	1.08	24.45	6.08	6.13	0.41	41.18
Prime × Moderate	1.27	2.60	20.19	3.65	15.47	0.92	39.54
Prime × High	1.54	2.83	19.70	2.55	21.35	1.02	39.68
Near-Prime × Low	3.20	9.30	20.38	8.60	21.66	3.45	40.05
Near-Prime × Moderate	4.55	21.05	14.89	7.31	32.57	5.61	38.35
Near-Prime × High	4.33	19.11	14.34	5.80	34.54	5.04	37.56
Subprime × Low	8.04	36.79	17.87	9.80	45.53	12.63	39.70
Subprime × Moderate	9.99	62.31	14.56	6.19	58.70	17.96	37.10
Subprime × High	7.65	58.46	11.50	3.14	59.69	12.19	35.72

Note. Five-percent operational subsample, pooled over the four test windows; the sentinel cell is omitted. Budget is the share of loan-months not automatically cleared.

Mondrian's budget falls with the LTV bucket inside every FICO tier while its precision rises with it, so its largest allocations sit where its queue is thinnest in delinquencies. Its precision spans 53.6 points across the grid against SCP's 13.0 and APS's 5.5. The two global-threshold methods order the cells by budget identically but for the Subprime × Low and Subprime × High pair, and both track the realised positive rate. APS's ordering reproduces it exactly, while SCP's differs only in that same pair, though that rate is not monotone in the LTV bucket within the near-prime and subprime tiers.

G.2 Monthly Review-Workload Stability

Table G.2 summarises the monthly series behind the volatility comparison of §4.4, over the 72, 81, 19 and 24 observed months of the four test windows.

Table G.2: Monthly review budget by method and regime

Method	Regime	Months	Mean (%)	SD (pp)	Min (%)	Max (%)	CV
SCP	Crisis	72	8.53	0.99	6.74	9.84	0.116
	Normal	81	4.84	0.84	3.89	6.65	0.174
	COVID	19	4.32	0.61	3.48	5.94	0.140
	Rate hike	24	3.08	0.16	2.88	3.50	0.051
Mondrian	Crisis	72	9.39	0.93	7.18	10.68	0.099
(FICO $\times$ LTV)	Normal	81	5.05	0.75	4.31	7.05	0.148
	COVID	19	5.18	0.57	4.39	6.61	0.109
	Rate hike	24	4.16	0.31	3.83	4.95	0.075
APS	Crisis	72	2.55	0.33	1.94	3.05	0.129
	Normal	81	1.78	0.30	1.35	2.44	0.169
	COVID	19	1.52	0.36	1.09	2.62	0.236
	Rate hike	24	1.03	0.06	0.94	1.20	0.062

Note. Five-percent operational subsample; months are weighted equally and CV is SD divided by the mean.

Equal monthly weighting moves the means from the pooled budgets of Table G.4 by at most 0.04 percentage points. Mean within-regime standard deviations are 0.65, 0.64 and 0.26 points for SCP, Mondrian and APS. That ordering is scale-dependent and does not survive normalisation. On the coefficient of variation APS is the least stable of the three in the crisis and COVID windows and second in the other two. All three COVID series peak in one month, April 2020, at 5.94%, 6.61% and 2.62%. This carries APS's coefficient to 0.236.

G.3 Absolute Review and Capture Volumes

Table G.3 extrapolates the five-percent operational subsample to full test windows of 633,244,002, 753,732,904, 201,447,058 and 306,340,880 loan-months.

Table G.3: Reviewed loan-months and captured delinquent loan-months by regime and method

Regime	Method	Reviewed (m)	Captured (m)	Captured less APS (000s)
Crisis	SCP	53.93	9.958	+3,009.4
	Mondrian	59.22	8.801	+1,852.5
	APS	16.23	6.948	—
Normal	SCP	36.24	6.709	+1,676.2
	Mondrian	37.67	6.090	+1,057.3

Table G.3, continued

Regime	Method	Reviewed (m)	Captured (m)	Captured less APS (000s)
	APS	13.35	5.033	—
COVID	SCP	8.66	1.603	+562.2
	Mondrian	10.39	1.538	+497.4
	APS	3.04	1.041	—
Rate hike	SCP	9.44	1.619	+504.0
	Mondrian	12.76	1.550	+435.8
	APS	3.15	1.115	—

Note. Reviewed excludes escalation; captured includes it (Table G.6).

The crisis comparison of §4.4 reads off the first block: SCP reviews 37.70 million more loan-months than APS for 3.01 million more delinquent loan-months. Mondrian reviews 42.99 million more for 1.85 million more.

G.4 Matched-Budget Capture Results

Table G.4 gives the comparison behind (3.8) in full. The budget is the non-auto-cleared share (§3.6), which for Mondrian exceeds its review rate by the escalated share of Table G.6. The baseline column is the capture a descending-$\hat{p}$ ranking achieves on the same number of loan-months.

Table G.4: Capture against a calibrated-probability threshold at a matched review budget

Method	Regime	Budget (%)	Conformal (%)	Baseline (%)	Δ_{cap} (pp)
SCP	Crisis	8.52	62.05	62.05	$+0.000$
	Normal	4.81	58.12	58.12	$+0.000$
	COVID	4.30	47.69	47.69	$+0.000$
	Rate hike	3.08	50.81	50.81	$+0.000$
Mondrian	Crisis	9.38	54.84	63.58	-8.739
(FICO $\times$ LTV)	Normal	5.02	52.76	58.58	-5.818
	COVID	5.16	45.76	49.50	-3.732
	Rate hike	4.17	48.66	54.09	-5.426
APS	Crisis	2.56	43.30	43.30	$+0.000$
	Normal	1.77	43.60	43.60	$+0.000$
	COVID	1.51	30.96	30.96	$+0.000$
	Rate hike	1.03	34.98	34.98	$+0.000$

Note. APS uses the deterministic $U = 1$ reduction of §3.5.

The zeros are structural. For SCP and deterministic APS, non-auto-clear is a one-sided threshold in $\hat{p}$, so matching the exact row budget selects the same upper-tail rows as the probability baseline. Mondrian's deficit is the price of the reordering whose cell-level form is Table G.1.

G.5 Operational Results Under the Logistic Benchmark

Table G.5 repeats Table G.4 on the isotonic-calibrated logistic benchmark, evaluated on the same test rows. That pipeline's coverage results are in Appendix E.11.

Table G.5: Matched-budget capture under the logistic benchmark

Method	Regime	Budget (%)	Conformal (%)	Baseline (%)	Δ_{cap} (pp)
SCP	Crisis	7.91	59.88	59.88	$+0.000$
	Normal	5.53	57.98	57.98	$+0.000$
	COVID	4.77	43.65	43.65	$+0.000$
	Rate hike	3.69	49.51	49.51	$+0.000$
Mondrian (FICO $\times$ LTV)	Crisis	8.51	53.51	60.87	-7.365
	Normal	5.21	53.14	57.17	-4.024
	COVID	5.68	45.09	46.85	-1.762
	Rate hike	4.93	49.13	53.37	-4.233
APS	Crisis	3.08	44.06	44.06	$+0.000$
	Normal	2.32	45.03	45.03	$+0.000$
	COVID	1.99	28.03	28.03	$+0.000$
	Rate hike	1.52	35.58	35.58	$+0.000$

Note. Five-percent benchmark sample (Appendix B.3); ACI, DtACI and the coarse Mondrian partitions were not carried over (Appendix D.4).

No budget differs from its primary-pipeline counterpart by more than 0.87 percentage points, the widest being Mondrian's in the crisis window. Mondrian's mean deficit is 4.346 points against 5.929 on the primary pipeline, the shallower figure accompanying the narrower cell-threshold spread recorded in Appendix E.11. The operating-point ordering also carries: APS runs the smallest queue at the highest within-budget delinquency rate, 23.4% to 36.2%, and Mondrian the lowest of the three in every window.

G.6 Prediction-Set Composition and Routing Semantics

Escalation under the disposition mapping of §2.6 requires the delinquent label to be the only one retained, so it is governed by the upper cut-point. Table G.6 gives the calibrated probability at which each method crosses that bar and the share of loan-months that do.

Table G.6: Escalation boundary and realised escalation rate by method and regime

Method	Escalation bar on $\hat{p}$	Escalated (%) Crisis	Normal	COVID	Rate hike
SCP	0.9838	0.0000	0.0000	0.0000	0.0000
Mondrian (FICO $\times$ LTV)	0.5799–0.9967	0.0327	0.0179	0.0047	0.0023
APS ($U = 1$)	0.9000	0.0003	0.0001	0.0000	0.0000

Note. Five-percent operational subsample. Bars are $1 - \hat{q}$ for SCP, $1 - \hat{q}_g$ for Mondrian (Appendix D.1), and $\hat{q}_{\text{APS}}$ for APS. A displayed 0.0000 denotes a rate below 5×10^{-5}%, not necessarily zero.

Capture and class-conditional coverage then differ by construction. The performing singleton serves neither, the positive singleton and the two-label set serve both, and the empty set counts towards capture alone, so within any one construction and sample

$$\text{Capture} - \widehat{\text{Cov}}(y=1) = \widehat{\mathbb{P}}\big(C(X) = \emptyset \mid Y = 1\big). \tag{G.1}$$

Under SCP and Mondrian, whose review set is the empty one, the right-hand side is essentially the whole of capture. Under the $U = 1$ reduction it is zero: the APS review set holds the delinquent label, so deterministic capture and deterministic class-conditional coverage are one number.

The APS rates of Table 4.2 are the randomised construction's, and the two are not two measurements of the same quantity. Dropping the $(1 - U)\hat{p}_y$ term of (3.3) can only admit labels, so the $U = 1$ sets contain the randomised ones and their coverage is weakly the larger. The distance from 17.15–29.57% to the 30.96–43.60% capture of Table 4.3 is that nesting. The same term sets the queue sizes: under randomisation APS routes to review the union of its empty and two-label sets, 10.16% to 10.75% of loan-months (Appendix E.1), where the reduction emits no empty set at all and leaves 1.03% to 2.56%.

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Tübingen, den 25.08.2026

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